KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

HOSPITAL CHATBOT FOR COLOR-CODED CLINICAL PATHWAYS

BY

QUAIGRAINE SAMUEL AND TWUM SAMUEL

A THESIS SUBMITTED TO THE DEPARTMENT OF MATHEMATICS, KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE OF BACHELOR OF SCIENCE IN MATHEMATICS

2026

DECLARATION

We hereby declare that this submission is our own work towards the award of the Bachelor of Science degree in Mathematics and that, to the best of our knowledge, it contains no material previously published by another person nor material which has been accepted for the award of any other degree of the university, except where due acknowledgment has been made in the text.

Quaigraine Samuel	....	....
Student	Signature	Date

Twum Samuel	....	....
Student	Signature	Date

Certified by:

Professor Peter Amoako Yirenkyi	....	....
Supervisor	Signature	Date

Certified by:

Professor Charles Sebil	....	....
Head of Department	Signature	Date

DEDICATION

To my family, lecturers, and the patients whose care motivates this work.

ABSTRACT

Triage is the structured process of assigning each arriving patient an urgency level so that hospital resources match clinical need. In low- and middle-income countries, including Ghana, overcrowding, delayed first assessment, and *acuity discontinuity* (urgency colour stops guiding care after intake) affect outpatient departments, wards, and emergency streams alike. This thesis develops a hospital chatbot that maps unstructured *chief complaints* (unstructured descriptions of why the patient has come) to South African Triage Scale (SATS) colours and clinical pathways.

The mathematical model converts English patient text to a colour through a fine-tuned Biomedical Bidirectional Encoder Representations from Transformers (BioBERT) classifier. Tokenisation, contextual embeddings, multi-head self-attention, and softmax classification are specified with full formulas. A fixed map f_{SATS} converts predicted acuity levels to Red, Orange, Yellow, or Green. The Triage Early Warning Score (TEWS) is presented as the parallel hospital standard for objective vital signs; when vitals are unavailable at intake, the text model supplies the initial colour. A clinical relevance gate filters non-medical input before classification.

Holdout evaluation on the 80,000-pair Triagegeist benchmark reports 99.92% accuracy. Synthetic Minority Over-sampling Technique (SMOTE) and stratified training mitigate class imbalance, raising Level 1 (Red) recall from 98.14% to 100% at a small precision cost, prioritising a safety-first clinical trade-off.

The methodology is illustrated on a running complaint at KNUST University Hospital: *"I have a headache and feel feverish. My body aches and I feel weak."* System architecture and the prototype `/predict` application programming interface (API) are documented in Chapter 3 and Appendix 5.2. Results and conclusion are reported in later chapters.

ACKNOWLEDGMENT

We thank our supervisor, Professor Peter Amoako Yirenkyi, and the Department of Mathematics at Kwame Nkrumah University of Science and Technology for guidance throughout this project. We also acknowledge the open-source communities behind BioBERT, Hugging Face Transformers, and the clinical triage literature that informed the mathematical model developed here.

CONTENTS

CHAPTER 1

INTRODUCTION

1.1 Background

Hospitals in Ghana operate under sustained pressure: crowded waiting areas, limited nursing staff, and long queues before a clinician sees a new arrival. Triage is the mechanism that aligns patient urgency with resource allocation. In well-resourced systems, urgency is reassessed as patients move through diagnostics, wards, and outpatient streams. In resource-constrained settings, however, standardised protocols such as the South African Triage Scale (SATS) are often applied only at the first desk. Once a patient leaves that point, their colour-coded acuity may no longer steer bed allocation, laboratory orders, or re-assessment timers. We call this failure *acuity discontinuity*.

Teaching hospitals in Ghana, including KNUST University Hospital, apply SATS to structure intake across clinical departments, not only in a single emergency bay. Colour codes (Red, Orange, Yellow, Green, and Blue for dignity protocols) express how quickly a patient must be seen and where they should stream next: resuscitation, acute beds, urgent waiting, or routine outpatient care. Nurses at triage also compute a Triage Early Warning Score (TEWS) from objective vital signs. Yet many delays begin earlier, when a patient arrives with only words: a headache, fever, weakness, or abdominal pain described in free text before any thermometer or pulse oximeter reading is available.

This thesis proposes a digital *Flow Management Engine* in the form of a hospital chatbot. The chatbot accepts a single English free-text *chief complaint* (the primary reason the patient presents, in their own words; see Section 3.1.1) and returns a SATS colour reflecting urgency together with acuity probabilities and a clinical pathway card that can guide routing across outpatient departments, wards, and intake queues. The core contribution is mathematical: a pipeline from unstructured complaint text to acuity probabilities, colour, and pathway card, developed in Chapter 3.

1.2 Problem statement

Low- and middle-income country (LMIC) hospitals struggle with crowded waiting rooms and limited human resources at intake. Patients may wait hours before first assessment. Critical cases are not always identified quickly, while less urgent presentations congest acute zones. Medical systems are confusing; without immediate direction, patients do not know whether to join the emergency stream, the polyclinic, or self-care advice.

Manual, paper-based triage creates acuity discontinuity: patient status becomes static rather than dynamic. Two systemic failures follow.

1. **Lost urgency.** Patients triaged Orange or Red can lose time-critical priority when they move to imaging, laboratories, or wards, because no digital record preserves their colour and countdown timer.
2. **Green congestion.** Non-urgent (Green) patients lack automated guidance toward outpatient or self-care routes, blocking beds and staff needed for life-saving care.

Without a framework to manage flow hospital-wide, overcrowding and resource mismatch persist; the most vulnerable patients become invisible to the system after the first desk.

1.3 Objectives

1. **Automate triage classification.** Develop a chatbot that classifies a single unstructured chief complaint submitted by the patient or attendant, producing acuity probabilities, a SATS colour, and a pathway card reflecting clinical urgency.
2. **Guide clinical pathways.** Provide immediate, actionable next-step guidance: critical (Red) cases toward urgent care; non-urgent (Green) cases toward self-care or routine clinic streams.
3. **Integrate with hospital protocols.** Align decision logic with SATS and evidence-based pathway tables so recommendations remain safe and auditable.
4. **Collect anonymised data for refinement.** Use de-identified intake records to improve the classification model and to inform administrators about patient flow and common presentations.

CHAPTER 2

LITERATURE REVIEW

2.1 Hospital overcrowding and triage delay

Overcrowding in low- and middle-income country (LMIC) hospitals affects outpatient departments, emergency intake, and inpatient wards simultaneously (KATH Emergency Department, 2018). Delayed triage lengthens time-to-provider, increases left-without-being-seen rates, and correlates with poor outcomes for time-sensitive conditions. The South African Triage Scale (SATS) provides a five-colour framework (Red, Orange, Yellow, Green, Blue) with maximum waiting times $T_{\max}$ and destination rules (South African Triage Group, 2012). Implementation at Komfo Anokye Teaching Hospital (KATH) demonstrated that standardised colour coding can structure flow even when staffing is constrained, provided urgency information travels with the patient.

A persistent gap is the *text-first* moment of arrival: many patients can describe symptoms before any vital sign is measured. Paper triage forms capture chief complaints as unstructured prose; converting that prose into the same colour logic used for vitals remains inconsistently formalised in Ghanaian hospital practice.

2.2 TEWS and vital-sign triage

The Triage Early Warning Score (TEWS) aggregates six SATS parameters (heart rate (HR), respiratory rate (RR), mobility, temperature, Alert, Voice, Pain, Unresponsive (AVPU) consciousness, and trauma flag) into a single integer T (South African Triage Group, 2012). Nurses use T at triage to assign a colour band: Red for $T > 7$, Orange for $5 \leq T \leq 6$, Yellow for $3 \leq T \leq 4$, Green for $T \leq 2$. TEWS is deterministic and objective; it does not read language. Subjective symptoms such as "worst headache of my life" or "feverish and weak" may appear in the complaint before corresponding vitals exist. Chapter 3 presents TEWS mathematics as the hospital standard parallel to the text model.

2.3 Attention, BERT, and BioBERT

Sequential language models read text in one direction, limiting context for ambiguous clinical phrases. Vaswani et al. (2017) introduced scaled dot-product self-attention, allowing each token to attend to all others in parallel. Devlin et al. (2019) built BERT (Bidirectional Encoder Representations from Transformers) on this idea: masked language modelling (MLM) pre-training yields contextual embeddings $E(t_i) = E_{\text{word}} + E_{\text{pos}} + E_{\text{seg}}$ used across downstream tasks.

Lee et al. (2020) continued pre-training on PubMed and PMC abstracts to produce Biomedical BERT (BioBERT), a domain-specific encoder whose embedding space places “headache”, “feverish”, and “malaria” nearer clinically related terms. Fine-tuning BioBERT on nurse-labeled chief complaints yields a supervised classifier from text to acuity levels.

2.4 NLP for emergency and hospital triage

Stewart et al. (2023) reviewed natural language processing (NLP) at emergency triage: chief-complaint classification, syndromic surveillance, and decision support remain active research areas with heterogeneous datasets. Gligorijević et al. (2018) showed that deep attention models can stratify emergency department patients from unstructured text. Most published systems target high-income corpora; formal mathematics from patient text to SATS colour for Ghanaian hospital intake, with pathway routing aligned to local implementation, is under-specified in the literature.

2.5 Research gap

Existing work separates vital-sign scores (TEWS), manual SATS colour assignment, and NLP classifiers trained abroad. There is no unified, examinable mathematical account of how a free-text complaint at KATH-style intake becomes a SATS colour and pathway card using BioBERT, while TEWS remains documented as the nurse-facing vital standard. Chapter 3 fills that gap for the text path; later chapters will report evaluation.

CHAPTER 3

METHODOLOGY

This chapter specifies the mathematical pipeline from an arbitrary chief complaint to a South African Triage Scale (SATS) colour and clinical pathway. Where concrete numbers aid understanding, we refer to a recurring illustrative example at KNUST University Hospital: *"I have a headache and feel feverish. My body aches and I feel weak."* The same steps apply to any valid clinical text that passes the relevance gate in Section 3.16.

The chatbot supports triage staff; it does not diagnose disease or replace clinical judgment.

3.1 Methodological overview

The hospital chatbot draws on two complementary ideas from clinical practice; this thesis develops the text-based path in full and documents the Triage Early Warning Score (TEWS) as the parallel vital-sign standard.

First, **clinical pathways** encode hospital protocols: symptoms and context map to an urgency colour and a destination (resuscitation bay, acute bed, urgent waiting, outpatient stream). Pathways are the product the chatbot shows to patients and staff after classification.

Second, TEWS is the nurse-facing vital-sign score used at SATS triage desks. When heart rate (HR), respiratory rate (RR), mobility, temperature, Alert, Voice, Pain, Unresponsive (AVPU) consciousness, and trauma status are measured, TEWS produces a deterministic integer T and colour band $C_{\text{TEWS}}(T)$. TEWS is explained fully in Section 3.3 because it is the objective hospital standard; the chatbot's primary engine for *free-text chief complaints* at intake is a deep attention model (Biomedical BERT, BioBERT), not a weighted sum of vitals.

When vitals are not yet available, the text model below supplies the intake colour. Nurses may later record vitals and apply TEWS independently; this thesis specifies the natural language processing (NLP) path only.

3.1.1 Chief complaints

The *chief complaint* is the primary reason the patient presents, expressed in their own words (or an attendant's) as unstructured free text at intake, before vitals, physical examination, or formal diagnosis. Nurses and triage staff traditionally record this as prose on paper or electronic forms; it captures symptom onset, severity, and context (for example, "headache and feverish, body aches, feel weak"). It is not a diagnosis, not a TEWS vital-sign score, and not a full medical history: it is only the presenting problem text X that this thesis classifies.

The chatbot accepts one **English** chief-complaint string per submission. The running example *"I have a headache and feel feverish. My body aches and I feel weak."* illustrates a typical Yellow-level presentation used throughout this chapter; pipeline symbol X appears in Table 3.1 and Figure 3.1.

3.1.2 Training data provenance

Model training and holdout evaluation draw on two distinct data layers.

Training and evaluation corpus. Stage 3 fine-tuning uses the *Triagegeist* synthetic emergency-department benchmark released by the Laitinen-Fredriksson Foundation on Kaggle (Laitinen-Fredriksson Foundation, 2025). Structured intake records (`train.csv`, acuity labels) are inner-joined with free-text rows (`chief_complaints.csv`) on `patient_id`, yielding $N = 80{,}000$ English `chief_complaint_raw` / `triage_acuity` pairs. Vitals and demographics in the source files are excluded so the NLP path matches single-shot text input at intake (Section 3.13.3). Five-level competition labels map to SATS colours via f_{SATS}. A stratified 90/10 split (approximately 72,000 train / 8,000 holdout, seed 42) supplies all reported accuracy and recall figures in Chapter 4. This corpus was **not** collected at KNUST University Hospital.

KNUST-informed contextual examples. Following a site visit to KNUST University Hospital, the authors generated supplementary English chief complaints based on observations of intake workflow, SATS colour assignment, and department structure. These examples test the prototype chatbot, validate pathway routing (Section 3.15), and illustrate the pipeline (for example, *"I have a headache and feel feverish. My body aches and I feel weak."* in Table 3.20); this supports qualitative analysis only. They are **not** merged into the 80,000-pair fine-tuning corpus. Prospective collection of nurse-labeled complaints at KNUST University Hospital for model retraining remains future work (Chapter 5).

3.1.3 BERT and bidirectional context

Older language models read text sequentially (left-to-right or right-to-left). Vaswani et al. (2017) showed that *self-attention* lets every token attend to every other token in one pass. Devlin et al. (2019) built Bidirectional Encoder Representations from Transformers (BERT) on this mechanism: during pre-training, random tokens are masked and predicted from full left *and* right context via masked language modelling (MLM). The result is contextual embeddings: the vector for "feverish" differs depending on neighbouring words such as "headache" or "not".

3.1.4 BioBERT for biomedical text

General BERT embeddings under-represent clinical terms. Lee et al. (2020) continued MLM pre-training on PubMed and PMC, then released BioBERT with the same transformer skeleton as BERT but an embedding space tuned for biomedical language. For triage, BioBERT is fine-tuned on the Triagegeist corpus described in Section 3.1.2: supervised learning maps English complaint text X to acuity level probabilities $\hat{\mathbf{y}}$, then to SATS colour via a fixed map f_{SATS}.

3.1.5 The `[CLS]` summary token

Classification does not read all token outputs separately. An artificial token `[CLS]` is inserted at position $i = 0$. After $L = 12$ encoder layers, its hidden state $h_{[\text{CLS}]} \in \mathbb{R}^{768}$ summarises the entire complaint and feeds the softmax classification head (Section 3.7). The `[CLS]` row accumulates context from every other token through the stacked encoder layers.

3.1.6 Pipeline summary

Figure 3.1 previews the end-to-end natural language processing (NLP) path from a free-text chief complaint to a SATS colour and pathway card. Each box corresponds to one transformation; the subsections and sections that follow derive the formulas, symbols, and worked steps for every stage.

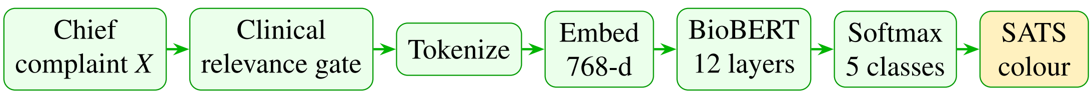

Figure 3.1: NLP pipeline from chief complaint to SATS colour (TEWS is documented separately in Section 3.3).

1. **Chief complaint X.** The patient or attendant submits one unstructured **English** text string describing symptoms and relevant history (Section 3.1.1). Tokenisation begins in Section 3.6.

2. **Clinical relevance gate.** A rule-based score $g(X)$ filters greetings and non-medical text before BioBERT runs; see Section 3.16.

3. **Tokenize.** WordPiece splits X into subword tokens with special markers `[CLS]`, `[SEP]`, and `<PAD>`; see Section 3.6.

4. **Embed (768-d).** Each token ID maps to a 768-dimensional vector; positional and segment embeddings are added to form the input matrix $H^{(0)}$; see Sections 3.8 and 3.9.

5. **BioBERT (12 layers).** Twelve encoder layers apply multi-head self-attention and feed-forward transforms; see Sections 3.10 and 3.11.

6. **Softmax (5 classes).** The `[CLS]` summary vector $h_{[\mathrm{CLS}]}$ passes through a classification head yielding acuity probabilities $\hat{\mathbf{y}}$; see Section 3.12.

7. **SATS colour and pathway.** The predicted level $\hat{c}$ maps to colour $C_{\mathrm{NLP}} = f_{\mathrm{SATS}}(\hat{c})$ and pathway card $P(C_{\mathrm{NLP}})$; see Sections 3.14 and 3.15.

Each stage above is derived and exemplified in the referenced sections. Section 3.17 restates the full path for the implemented prototype.
Table 3.1 records the mathematical object entering and leaving each stage so the reader can follow how one complaint string becomes a colour and pathway card.

Stage	**Input**	**Output**
Chief complaint	Patient text string	X (raw string)
Clinical gate	X	Pass/fail via $g(X)$
Tokenize	X	Token sequence $(t_1, \dots, t_M)$, $M \le 128$
Embed	Token IDs	Input matrix $H^{(0)} \in \mathbb{R}^{M \times 768}$
BioBERT	$H^{(0)}$	Final matrix $H^{(12)}$; summary $h_{[\mathrm{CLS}]}$
Softmax	$h_{[\mathrm{CLS}]}$	Acuity probabilities $\hat{\mathbf{y}} \in \mathbb{R}^5$
SATS colour	$\hat{\mathbf{y}}$	Level $\hat{c}$, colour C_{NLP}, pathway $P(C_{\mathrm{NLP}})$

Table 3.1: Pipeline stage inputs and outputs (NLP path).

3.2 SATS colour chart

SATS assigns each living patient an urgency colour with a maximum waiting time $T_{\max}$ and hospital destination (South African Triage Group, 2012). Figure 3.2 separates immediate and special protocols from the decreasing-urgency scale used for living patients. Blue indicates deceased-on-arrival dignity protocol, not low urgency.

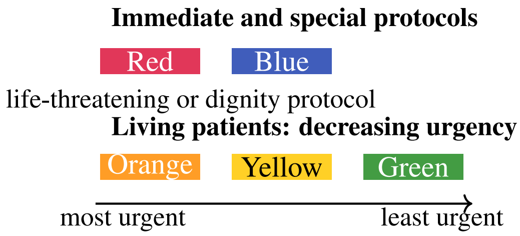

Figure 3.2: South African Triage Scale (SATS) colour bands used at hospital intake.

Colour	**Meaning**	$T_{\max}$	**Typical destination**
Red	Immediate life threat	0 min	Resuscitation bay
Orange	Very urgent	10 min	Acute / high-dependency bed
Yellow	Urgent	60 min	Emergency waiting / urgent stream
Green	Non-urgent	$<$ 4 h	Outpatient department (OPD) / minors / polyclinic
Blue	Dignity protocol	N/A	Mortuary / private space

Table 3.2: SATS colours, waiting targets, and routing (SATS-aligned).

Remark 3.1. *Red signals an immediate threat to life; Green indicates that the patient may safely wait in a routine stream. The chatbot outputs one of these colours so intake staff can route patients without re-interpreting free text.*

For the illustrative complaint in this chapter, the NLP model in Section 3.12 may assign moderate urgency (Yellow) with high probability; the same framework applies regardless of presenting symptoms.

3.3 TEWS: Triage Early Warning Score

TEWS is the deterministic score nurses compute from six vital parameters at SATS triage (South African Triage Group, 2012). It provides an objective colour band when measurements exist.

3.3.1 Definition

Let $\mathbf{v} = (v_1, \ldots, v_6)$ denote HR, RR, mobility, temperature, AVPU consciousness, and trauma flag. Each parameter maps to points $f_k(v_k) \in \{0, 1, 2, 3\}$ via SATS lookup tables. TEWS aggregates these contributions into one integer so that nurses at different desks can compare patients on a common scale. Weights $w_k = 1$:

$$T = \sum_{k=1}^{6} w_k f_k(v_k) = \sum_{k=1}^{6} f_k(v_k). \tag{3.1}$$

Summing six bounded terms yields a score $T \in [0, 18]$ in principle, though typical values at triage are much lower. Because each f_k is defined from the SATS manual, the score is reproducible and auditable: two nurses who measure the same vitals obtain the same T.

k	Parameter v_k	Source	Unit / scale
1	Heart rate (HR)	SATS manual, TEWS table	beats/min
2	Respiratory rate (RR)	SATS manual, TEWS table	breaths/min
3	Mobility	SATS manual	walking / assisted / immobile
4	Temperature	SATS manual	°C
5	AVPU consciousness	SATS manual	alert / verbal / pain / unresponsive
6	Trauma	SATS manual	present / absent

Table 3.3: TEWS parameters and documentation source.

Remark 3.2. *Each vital sign contributes zero to three points. Elevated HR, abnormal RR, or reduced consciousness increase T; normal values contribute zero.*

3.3.2 Piecewise scoring functions

The piecewise functions f_1 and f_2 encode tachycardia, bradycardia, and abnormal respiratory rate as higher point values, reflecting physiological stress seen at triage.
Heart rate f_1 (beats/min):

$$f_1(v_1) = \begin{cases} 3 & v_1 \geq 130 \\ 2 & 111 \leq v_1 \leq 129 \text{ or } v_1 \leq 40 \\ 0 & 51 \leq v_1 \leq 100 \\ 1 & \text{otherwise} \end{cases} \tag{3.2}$$

Elevated heart rate (tachycardia) or very low heart rate therefore increases T and may raise the TEWS colour band.

Respiratory rate f_2 (breaths/min):

$$f_2(v_2) = \begin{cases} 3 & v_2 \geq 30 \\ 2 & 21 \leq v_2 \leq 29 \\ 0 & 9 \leq v_2 \leq 14 \\ 1 & \text{otherwise} \end{cases} \tag{3.3}$$

Rapid breathing (tachypnoea) similarly adds points, capturing respiratory distress before full clinical assessment.

Mobility f_3: walking $= 0$; assisted $= 1$; immobile $= 2$–3 (SATS manual). Temperature f_4: 35.0–38.4 °C $= 0$; borderline fever or hypothermia $= 1$; extreme values $= 2$–3. AVPU f_5: alert $= 0$, verbal $= 1$, pain $= 2$, unresponsive $= 3$. Trauma f_6: none $= 0$; major trauma $= 2$–3.

3.3.3 Colour map

The mapping $C_{\text{TEWS}}(T)$ converts the integer score into the same colour vocabulary used at the SATS desk, so vital-sign triage aligns with hospital routing rules (South African Triage Group, 2012):

$$C_{\text{TEWS}}(T) = \begin{cases} \text{Red} & T > 7 \\ \text{Orange} & 5 \leq T \leq 6 \\ \text{Yellow} & 3 \leq T \leq 4 \\ \text{Green} & T \leq 2 \end{cases} \tag{3.4}$$

Thresholds in Equation (3.4) follow the standard SATS implementation: a single point difference can change the colour band, so nurses record each f_k carefully. The diagram below visualises these bands on the score axis.

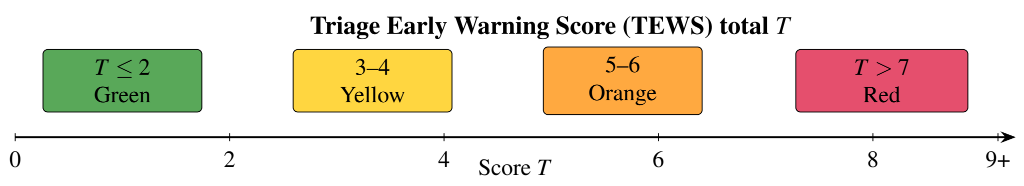

Figure 3.3: TEWS score ranges mapped to SATS colours (segments separated for readability).

k	Parameter	Measured v_k	$f_k(v_k)$	Running sum
1	HR	125 bpm	2	2
2	RR	26 breaths/min	2	4
3	Mobility	walking	0	4
4	Temperature	36.8 °C	0	4
5	AVPU	alert	0	4
6	Trauma	absent	0	4
			Total T:	**4**

Table 3.4: Term-by-term TEWS calculation for HR = 125 bpm, RR = 26 breaths/min, all other parameters normal.

Example 3.1. *Suppose vitals are HR* $= 125$ *bpm and RR* $= 26$ *breaths/min; mobility, temperature, AVPU, and trauma are normal or absent. Table 3.4 lists each* f_k *contribution; summing gives*

$$\begin{aligned}
f_1(125) &= 2 && (\textit{HR band } 111\text{–}129),\\
f_2(26) &= 2 && (\textit{RR band } 21\text{–}29),\\
f_k(\cdot) &= 0 && k = 3,\ldots,6,\\
T &= 2+2 = 4, &&\\
C_{\text{TEWS}}(4) &= \textit{Yellow} \quad (\textit{Equation } (3.4)). &&
\end{aligned}$$

If only these two vitals were measured, the score is partial but still auditable. At intake, before vitals exist, the BioBERT path in Section 3.12 supplies the text-based colour.

3.4 Introducing BERT

TEWS ignores language. A patient may describe symptoms in free text before HR or RR are recorded. Subjective complaints often appear first; bidirectional encoding is required so that “not feverish” and “very feverish” are not treated alike.
Vaswani et al. (2017) introduced scaled dot-product attention (Section 3.11). Devlin et al. (2019) stacked attention into deep encoders with MLM pre-training. The input representation preview is

$$E(t_i) = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(t_i), \tag{3.5}$$

where E_{word} identifies the token, E_{pos} encodes position i, and E_{seg} marks sentence segment (always segment 0 for a single complaint). Full detail appears in Section 3.8.

Term	Meaning	Example in triage
E_{word}	Token identity: which word or subword?	"headache" vs "feverish"
$E_{\text{pos}}(i)$	Position in the sequence: order matters	"head ache" $\neq$ "ache head"
E_{seg}	Sentence segment (A or B in paired inputs)	Always segment 0 for one complaint

Table 3.5: Breakdown of the three terms in Equation (3.5).

Remark 3.3. *BERT processes the full sentence simultaneously. The representation of "feel feverish" depends on "headache" and neighbouring tokens in both directions, which is essential for triage wording.*

3.5 Introducing BioBERT

BioBERT reuses BERT's architecture (12 layers, 768 hidden dimensions, 12 attention heads) but continues MLM on biomedical corpora. Fine-tuning on nurse-labeled chief complaints adjusts the classification head so $h_{[\text{CLS}]}$ maps to five acuity levels.
Clinical tokens such as "headache", "feverish", "aches", and "weak" occupy regions of embedding space nearer related medical terms than in general BERT. After fine-tuning, their joint attention pattern supports acuity estimation for any presenting complaint, not only the illustrative example used in this chapter.

Remark 3.4. *Hospital intake is frequently text-first. BioBERT provides the mathematical bridge from unstructured complaint text to the same colour vocabulary nurses use with TEWS and SATS.*

3.6 Tokenisation

Neural networks require fixed-size numerical input. Tokenisation is the first transformation: it maps raw complaint text X into a bounded sequence of subword tokens. The maximum length $M \leq 128$ matches the BioBERT configuration used at inference and allows batching of complaints of different lengths by padding shorter sequences.

$$X \xrightarrow{\tau} (t_1, \ldots, t_M), \quad M \leq 128. \tag{3.6}$$

WordPiece is the subword algorithm used by BERT and BioBERT. Common clinical words ("fever", "pain") remain whole; rare or misspelled forms split into pieces (e.g. "headache" $\rightarrow$ `head+##ache`). This keeps the vocabulary finite while still representing novel spellings. Vocabulary $\mathscr{V}$ has $|\mathscr{V}| \approx 28{,}996$ entries; each token ID indexes a row in the embedding table. When a word splits into two subwords, it occupies **two positions** in the sequence (see rows 4 and 7 in Table 3.7), so M counts subword tokens, not raw words.

Remark 3.5. *The fixed length $M = 128$ is a practical choice: BioBERT was trained with this cap, and padding shorter complaints to 128 lets the hospital batch many requests through the same matrix operations at intake.*

Three special tokens structure every input sequence:

- `[CLS]` (ID 101) is inserted at position $i = 0$ as the classification summary token; its final hidden state feeds the acuity head (Section 3.7).
- `[SEP]` (ID 102) is the **separator** token marking the end of the complaint sentence. It tells the model where clinical text stops, so subsequent `<PAD>` fillers are not interpreted as symptoms.
- `<PAD>` (ID 0) fills unused positions up to length $M = 128$; the attention mask sets these positions to zero so they do not affect encoding.

Token	**ID**	**Purpose**
`[CLS]`	101	Classification summary; always first
`[SEP]`	102	End-of-sentence boundary before padding
`<PAD>`	0	Batch padding to fixed length 128

Table 3.6: Special tokens in the BioBERT tokenizer.

i	Raw	Token	ID
0	n/a	`[CLS]`	101
1	I	i	1045
2	have	have	2031
3	a	a	1037
4	headache	head+##ache	7994, 8772
5	and	and	1998
6	feel	feel	2514
7	feverish	fever+##ish	9643, 6804
8	.	.	1012
9	n/a	`[SEP]`	102
10–127	n/a	`<PAD>`	0

Table 3.7: Token IDs for the illustrative complaint (padding to length 128 omitted).

3.7 The `[CLS]` and `[SEP]` tokens

Sequence classification requires one fixed-length vector summarising the entire complaint. Rather than averaging all token rows (which would weight filler words

equally with symptoms), BERT introduces an artificial summary token [CLS] at position $i = 0$. The [SEP] token at the end of the real text marks the sentence boundary: encoding stops treating tokens after [SEP] as part of the clinical complaint, which matters once <PAD> fillers occupy the remaining positions. Only the [CLS] row is passed to the classification head; [SEP] itself is not used for acuity prediction.
After $L = 12$ encoder layers:

$$h_{[\mathrm{CLS}]} = H^{(L)}_{0,:} \in \mathbb{R}^{768}. \tag{3.7}$$

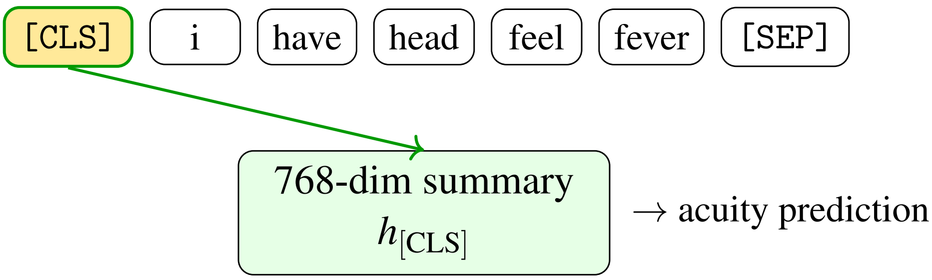

Figure 3.4: [CLS] aggregates context into one summary vector for classification.

Self-attention updates the [CLS] row at every layer so that it encodes information from all symptom tokens; the classification head therefore reads a single vector that represents the full clinical wording of X.

3.8 Embedding lookup

A token ID is an index, not a meaning. The embedding layer maps each ID to a dense vector in $\mathbb{R}^{768}$ via the learned matrix $E_{\mathrm{word}} \in \mathbb{R}^{V \times 768}$:

$$\mathbf{e}_i = E_{\mathrm{word}}[\mathrm{id}(t_i)] \in \mathbb{R}^{768}. \tag{3.8}$$

Symbol	Definition
V	Vocabulary size ($\approx$ 30,000)
$D = 768$	Hidden dimension (BioBERT base configuration)
$\mathbf{e}_i$	Word embedding vector for token at position i

Table 3.8: Embedding lookup notation.

The lookup $\mathbf{e}_i$ is one term in the full input representation $E(t_i) = E_{\mathrm{word}} + E_{\mathrm{pos}} + E_{\mathrm{seg}}$ from Equation (3.5). After PubMed pre-training, BioBERT places clinically related terms nearer in this space than general-purpose embeddings would.

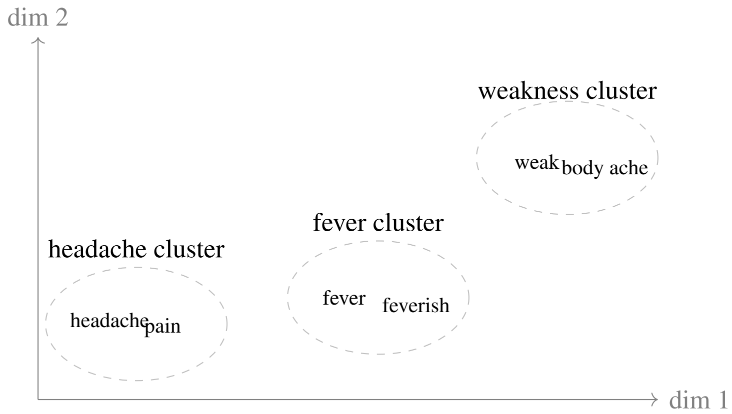

Figure 3.5: Schematic of BioBERT embedding space: familiar symptom words group together after biomedical pre-training (two examples per cluster).

When a patient writes “feverish” instead of “fever”, or “head ache” as two tokens, the model can still map the wording into regions of this space associated with similar presentations. That generalisation across varied patient language is central to text-first triage.

Contextual mapping sums three embeddings (Equation (3.5)). The positional embedding depends only on index i, and the segment embedding depends on segment id s (always $s = 0$ for a single complaint):

$$E_{\text{pos}}(i) \in \mathbb{R}^{768}, \qquad E_{\text{seg}}(s) \in \mathbb{R}^{768}, \quad s = 0. \tag{3.9}$$

The final input vector for token t_i at position i is the element-wise sum across all $D = 768$ dimensions:

$$E(t_i)_d = E_{\text{word}}(t_i)_d + E_{\text{pos}}(i)_d + E_{\text{seg}}(s)_d, \quad d = 1, \ldots, 768. \tag{3.10}$$

Symbol	**Definition**
$E_{\text{word}}(t_i)$	Row of E_{word} looked up by token ID
$E_{\text{pos}}(i)$	Learned vector for position i in the sequence
$E_{\text{seg}}(s)$	Learned vector for segment s (here $s = 0$)
$E(t_i)_d$	Component d of the summed embedding

Table 3.9: Symbols in the full input embedding (Equation (3.10)).

Example 3.2. *For token “head” at position $i = 4$, suppose the first three dimensions* *(illustrative only) are*

$$\begin{aligned}
E_{\text{word}} &= [0.31, -0.18, 0.52], \\
E_{\text{pos}}(4) &= [0.05, 0.02, -0.01], \\
E_{\text{seg}}(0) &= [0.00, 0.00, 0.00], \\
E(t_4) &= E_{\text{word}} + E_{\text{pos}}(4) + E_{\text{seg}}(0) \\
&= [0.36, -0.16, 0.51] \quad \textit{(first three dimensions)}.
\end{aligned}$$

The same element-wise addition repeats for all remaining dimensions $d = 4, \ldots, 768$. The result is row 4 of $H^{(0)}$.

Remark 3.6. *The embedding table is like a dictionary: each token ID points to a list of 768 numbers encoding meaning. Adding position tells the model where the word appears; adding segment (zero here) reserves the mechanism for two-sentence inputs in other BERT tasks.*

Figure 3.6 shows the flow for one token position. The 768 dimensions are a fixed width chosen in the original BERT architecture: wide enough to encode rich semantics, yet small enough to train twelve layers on large corpora.

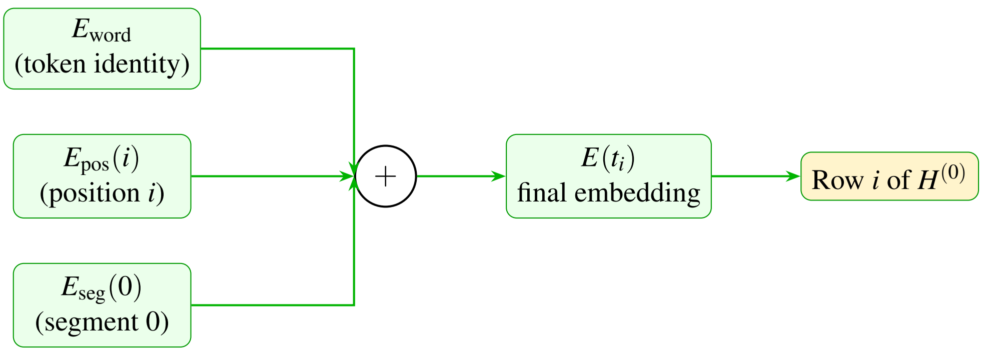

Figure 3.6: Three embeddings summed element-wise for one token position.

3.9 Input matrix $H^{(0)}$

Stacking the embedding vector for each of the M tokens forms the initial hidden-state matrix

$$H^{(0)} \in \mathbb{R}^{M \times 768}. \tag{3.11}$$

Row i is built by applying Equation (3.10) at position i: look up the word vector, add the positional vector $E_{\text{pos}}(i)$, add the segment vector $E_{\text{seg}}(0)$, and stack the result. Each row is one token; each column is one learned feature dimension. Viewed this way, a chief complaint is a table of numbers that the encoder layers progressively transform.

Remark 3.7. *Rows are tokens and columns are features. The matrix shape $M \times 768$ is preserved through all twelve encoder layers; only the entries change as self-attention distributes context across rows.*

i	**Token**	D_1	D_2	D_3	D_4	$\cdots$
0	`[CLS]`	0.02	0.11	-0.05	0.33	$\cdots$
1	i	-0.08	0.04	0.19	0.01	$\cdots$
2	have	0.15	-0.22	0.07	0.44	$\cdots$
4	head	0.31	-0.18	0.52	0.09	$\cdots$
7	fever	0.44	0.27	-0.11	0.61	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Table 3.10: Example submatrix of $H^{(0)}$ (first four of 768 dimensions; illustrative values).

After encoding, row 0 (`[CLS]`) holds the summary passed to the softmax classifier; the remaining rows are not used directly for the final colour decision.

3.10 Encoder layers

BioBERT applies the same encoder block $L = 12$ times. Each block first lets tokens attend to one another (mixing symptom phrases), then applies a feed-forward network (FFN) to each row independently. **Residual connections** add the block input back to its output so that gradients can flow through depth during training. **Layer normalisation** rescales activations after each sub-step for numerical stability. The update equations are

$$Z^{(\ell)} = \mathrm{LayerNorm}\big(H^{(\ell-1)} + \mathrm{MultiHeadAttention}(H^{(\ell-1)})\big), \tag{3.12}$$

$$H^{(\ell)} = \mathrm{LayerNorm}\big(Z^{(\ell)} + \mathrm{FFN}(Z^{(\ell)})\big), \tag{3.13}$$

$$\mathrm{FFN}(x) = W_2\,\mathrm{ReLU}(W_1 x + b_1) + b_2, \tag{3.14}$$

with $W_1 \in \mathbb{R}^{768\times3072}$, $W_2 \in \mathbb{R}^{3072\times768}$, and bias vectors $b_1 \in \mathbb{R}^{3072}$, $b_2 \in \mathbb{R}^{768}$ in the standard BERT-base configuration. The inner dimension 3072 is four times the hidden width 768, giving the FFN enough capacity to transform each token row nonlinearly.

Component	What it does
MultiHeadAttention	Twelve parallel self-attention heads; each learns different word relationships
FFN	Two linear layers with ReLU: expands each row to 3072 dims, then projects back to 768
LayerNorm	Rescales activations for stable training across depth
Residual (+)	Adds sub-layer input to output so earlier information is not lost

Table 3.11: Components inside one BioBERT encoder layer.

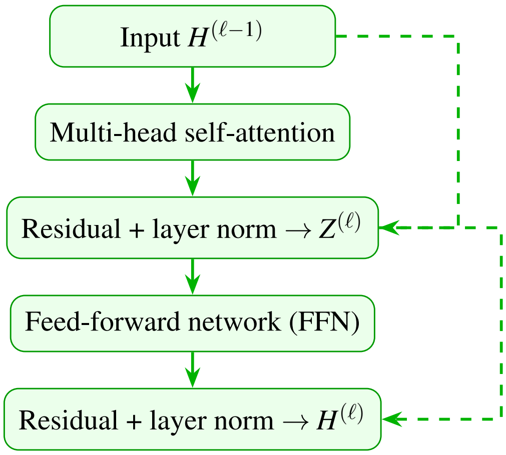

Figure 3.7: One BioBERT encoder layer (dashed arrows: residual connections). The block is stacked $L = 12$ times.

In the attention sub-step, each token row is updated using information from every other row in the complaint, which is how "headache" and "feverish" influence one another before acuity is estimated. The FFN sub-step then applies the same nonlinear transformation to each row separately, increasing representational capacity. Repeating this block twelve times transforms $H^{(0)}$ into $H^{(12)}$:

$$H^{(0)} \xrightarrow{\text{Layer 1}} H^{(1)} \xrightarrow{\text{Layer 2}} \cdots \xrightarrow{\text{Layer 12}} H^{(12)} = H^{(L)}.$$

Shallow layers (1–4) tend to encode individual words and short phrases; middle layers (5–8) combine symptoms such as "headache" with "feverish" and "weak"; deep layers (9–12) encode broader urgency patterns. All M token rows are processed in parallel at each layer, not left-to-right as in older language models.

Remark 3.8. *Stacking layers is like reading a complaint once for words, once for symptom combinations, and once for overall urgency. The `[CLS]` row at layer 12 becomes $h_{[\text{CLS}]}$ for classification.*

3.11 Self-attention

Self-attention computes a weighted combination of token representations for each position. The computation proceeds in four steps, matching Figure 3.8:

1. **Project** hidden rows into query, key, and value matrices.
2. **Score** pairwise compatibility between tokens.
3. **Normalise** scores into attention weights.
4. **Combine** value rows into context-aware outputs.

$$
\begin{aligned}
Q &= HW_Q, \quad K = HW_K, \quad V = HW_V, \\
S &= QK^\top / \sqrt{d_k}, \\
A &= \mathrm{softmax}(S), \\
\mathrm{output} &= AV.
\end{aligned}
$$

Following Vaswani et al. (2017), one attention head uses:

$$
\mathrm{Attention}(Q,K,V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{3.15}
$$

with key dimension $d_k = 64$ per head. For one query row, the softmax over keys assigns weight α_j to token j:

$$
\alpha_j = \frac{\exp(e_j)}{\sum_{k=1}^{m}\exp(e_k)}, \quad \sum_{j=1}^{m} \alpha_j = 1, \tag{3.16}
$$

where e_j is the scaled score in row j of $QK^\top/\sqrt{d_k}$ and m is the sequence length. BioBERT runs twelve such heads in parallel; their outputs are concatenated and projected back to 768 dimensions so each head can specialise (e.g. symptom co-occurrence vs negation).

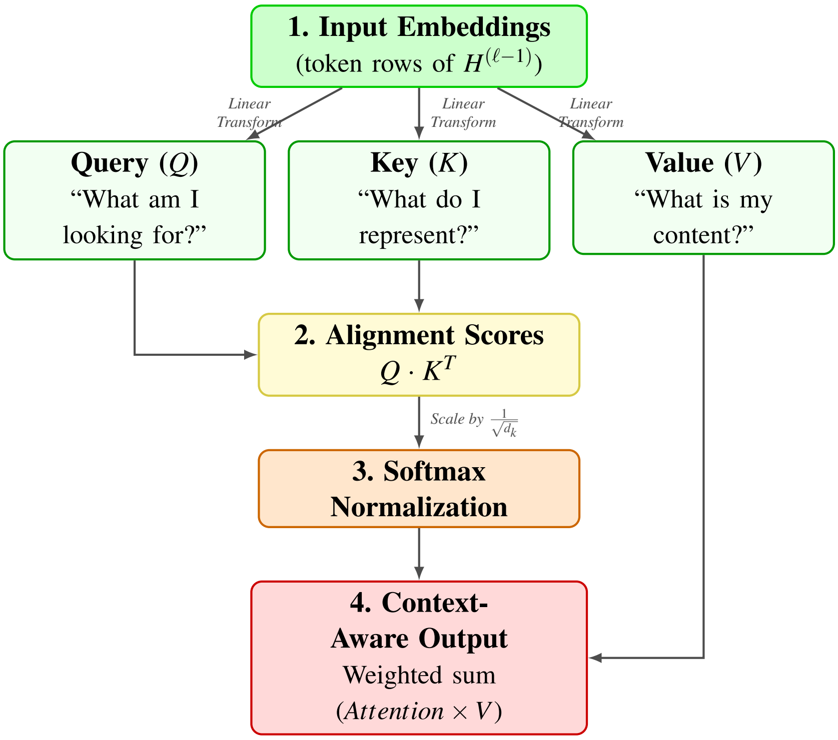

Figure 3.8: Self-attention data flow within one encoder layer.

Symbol	**Definition**
Q, K, V	Query, key, and value matrices derived from hidden states H
$QK^\top$	Matrix of pairwise compatibility scores between tokens
$\sqrt{d_k}$	Scaling constant ($d_k = 64$) that prevents dot products from growing too large
e_j	Raw attention score for token j before normalisation
α_j	Normalised attention weight for token j (a probability)
m	Sequence length (number of tokens, at most 128)

Table 3.12: Self-attention notation.

Query token	[CLS]	headache	feverish	filler
Attention to	0.09	0.32	0.41	0.18

Table 3.13: Schematic attention weights when the query is [CLS] (illustrative, not measured from the model). Symptom tokens receive higher weight than filler words.

The product $QK^\top$ measures how strongly each token should attend to every other token. Dividing by $\sqrt{d_k}$ keeps those scores in a range where the softmax in Equation (3.16) does not saturate. In a complaint mentioning both "headache" and "feverish", attention typically assigns higher weight to symptom tokens than to function words such as "I" or "and", so the updated representation emphasises clinically informative wording (Table 3.13).

3.12 Classification softmax

The classification head maps the summary vector $h_{[\mathrm{CLS}]}$ to five acuity logits, then applies softmax to obtain a probability distribution. Logits are unconstrained real scores; softmax converts them into non-negative probabilities that sum to one, which is the standard output format for multi-class clinical classifiers:

$$\begin{aligned} \mathbf{z} &= W h_{[\mathrm{CLS}]} + \mathbf{b}, \\ \hat{\mathbf{y}} &= \mathrm{softmax}(\mathbf{z}), \\ \hat{y}_c &= \frac{\exp(z_c)}{\sum_{j=1}^{5} \exp(z_j)}, \end{aligned} \tag{3.17}$$

with $W \in \mathbb{R}^{5\times 768}$, $\mathbf{b} \in \mathbb{R}^5$, logits z_c, and $\sum_{c=1}^{5} \hat{y}_c = 1$. The predicted level is $\hat{c} = \arg\max_c \hat{y}_c$. Expressing output as probabilities allows downstream routing logic to consider confidence (e.g. a dominant Yellow at 72% versus a near tie between Yellow and Green).

Symbol	Definition
$h_{[\mathrm{CLS}]}$	768-dimensional summary vector from final encoder layer
$W, \mathbf{b}$	Learned classification weight matrix and bias
z_c	Logit (pre-softmax score) for acuity level c
$\hat{y}_c$	Probability of level $c \in \{1, \dots, 5\}$
$\hat{c}$	Predicted level: index of largest $\hat{y}_c$

Table 3.14: Classification head notation.

Example 3.3. *Suppose illustrative logits for levels 1–5 are*

$$\begin{aligned} \mathbf{z} &= [-1.2, -0.5, 1.8, 0.3, -0.9], \\ \exp(\mathbf{z}) &\propto [0.30, 0.61, 6.05, 1.35, 0.41], \\ \sum_{c=1}^{5} \exp(z_c) &= 8.72, \\ \hat{\mathbf{y}} &\approx [0.03, 0.07, 0.69, 0.15, 0.05], \\ \hat{c} &= 3, \\ f_{\mathrm{SATS}}(3) &= \textit{Yellow}. \end{aligned}$$

Level 3 has the largest probability ($\hat{y}_3 \approx 0.69$, rounded to 72% in deployment logs; Section 3.14).

3.13 Model training pipeline

The deployed triage classifier is not trained from random initialisation. It is the product of three sequential stages: biomedical pre-training (Stage 1), encoder transfer from a prior triage checkpoint (Stage 2), and project fine-tuning on nurse-labeled chief complaints (Stage 3). Stages 1 and 2 supply medical language and partial triage adaptation; Stage 3 is the comprehensive retraining that maps free-text complaints to five acuity levels for this chatbot.

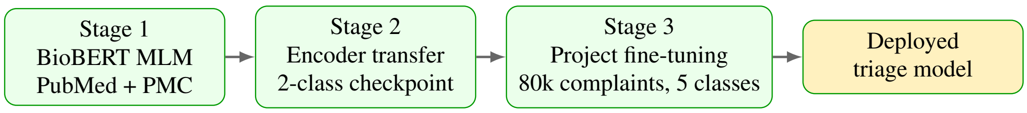

Figure 3.9: Three-stage training stack for the BioBERT triage classifier.

3.13.1 Stage 1: BioBERT pre-training (masked language modelling)

Before any triage labels are used, BioBERT is trained on biomedical text with masked language modelling (MLM). Random tokens are hidden and the model must predict them from surrounding context in both directions:

$$\mathscr{L}_{\mathrm{LM}} = -\sum_{i \in \mathscr{M}} \log P(t_i \mid H^{(L)}; \theta), \tag{3.18}$$

where $\mathscr{M}$ is the set of masked positions, t_i is the true token at position i, $H^{(L)}$ is the final hidden-state matrix from all twelve layers, θ denotes all model parameters, and $P(t_i \mid \cdot)$ is the softmax probability assigned to the correct word. PubMed abstracts and PubMed Central full text ($\approx$ 18 billion words) teach medical vocabulary and syntax. Stage 1 does **not** teach triage urgency, SATS colours, or informal chief-complaint phrasing.

Symbol	**Definition**
$\mathscr{M}$	Set of masked token positions in the sentence
t_i	True (correct) word at masked position i
θ	All model weights and biases
$P(t_i \mid H^{(L)}; \theta)$	Model probability for the correct token
$\mathscr{L}_{\mathrm{LM}}$	Total MLM penalty; lower means better predictions

Table 3.15: Masked language modelling notation (Stage 1).

Example 3.4. *If the masked token should be "headache", the MLM contribution is*

Good prediction ($P = 0.92$):

$$-\log(0.92) \approx 0.08 \qquad \textit{(small penalty)},$$

Poor prediction ($P = 0.01$):

$$-\log(0.01) \approx 4.6 \qquad \textit{(large penalty)}.$$

3.13.2 Stage 2: Encoder transfer from a triage checkpoint

A publicly available BioBERT triage checkpoint partially adapted the encoder to urgency-related text using a **two-class** classification head ($W \in \mathbb{R}^{2\times768}$). For this project, that 2-class head is **discarded** on load because our labels require five acuity levels. The loader uses `ignore_mismatched_sizes=True`: only encoder weights carry forward; the output layer is randomly reinitialised for five classes. Stage 2 therefore contributes triage-oriented encoder features without constraining the final colour mapping.

3.13.3 Stage 3: Fine-tuning on nurse-labeled chief complaints

Stage 3 is the comprehensive retraining performed for this thesis. Nurse-assigned acuity labels from the Triagegeist benchmark (Section 3.1.2) are inner-joined with chief-complaint text on patient identifier, yielding $N = 80{,}000$ English text-label pairs (X_i, y_i). Only `chief_complaint_raw` and `triage_acuity` are used; vitals and demographics in the source records are excluded so the NLP path matches the single-shot text input at intake.

The data are split 90/10 into training and validation sets ($\approx$ 72,000 / $\approx$ 8,000 rows, `test_size=0.1`, random seed 42). A new classification head $W \in \mathbb{R}^{5\times768}$, $\mathbf{b} \in \mathbb{R}^5$ is trained from scratch while encoder weights are updated with a small learning rate.

Level	**Count**	**Proportion**	**SATS colour**
1 (most urgent)	3,222	0.0403	Red
2	13,439	0.1680	Orange
3	28,921	0.3615	Yellow
4	23,020	0.2878	Green
5 (least urgent)	11,398	0.1425	Green
Total	**80,000**	**1.000**	

Table 3.16: Class distribution in the 80,000 labeled chief-complaint training pairs.

Hyperparameter	Value
Epochs	3
Batch size (train and eval)	16
Learning rate α	2×10^{-5}
Weight decay	0.01
Max sequence length	128 tokens
Mixed precision (FP16)	Enabled
Total optimizer steps	13,500 (3 epochs $\times$ 4,500 steps)
Best validation loss	0.001812 (epoch 3)

Table 3.17: Fine-tuning hyperparameters (Stage 3).

Component	Updated during Stage 3?
Embedding layers	Yes (small learning-rate updates)
12 transformer encoder layers	Yes
Classification head $W, \mathbf{b}$	Yes (trained from scratch)

Table 3.18: Which model components change during project fine-tuning.

The objective is **cross-entropy loss**: for each training example, the model is penalised when it assigns low probability to the nurse's true acuity label. For a single example with true class y,

$$\mathscr{L}_{\text{CE}} = -\log \hat{y}_y, \tag{3.19}$$

where $\hat{y}_y$ is the probability the model assigns to the correct level. Over the full training set:

$$\mathscr{L}_{\text{triage}} = -\sum_{i=1}^{N} \log P(y_i \mid X_i; \theta), \quad N = 80{,}000. \tag{3.20}$$

Example 3.5. *If the nurse labeled a complaint Yellow (level 3), the cross-entropy penalty is*

Low confidence on the true class ($\hat{y}_3 = 0.05$):

$$\mathscr{L}_{\text{CE}} = -\log(0.05) \approx 3.0 \qquad \textit{(large penalty)},$$

High confidence on the true class ($\hat{y}_3 = 0.95$):

$$\mathscr{L}_{\text{CE}} = -\log(0.95) \approx 0.05 \qquad \textit{(small penalty)}.$$

Fine-tuning adjusts weights so correct classes receive higher probability on similar complaints.

Backpropagation computes how each weight contributed to $\mathscr{L}_{\text{triage}}$ by applying the chain rule backward from the loss through the softmax head and all twelve encoder layers. For classification weights W, with logits $\mathbf{z} = Wh_{[\mathrm{CLS}]} + \mathbf{b}$,

$$\frac{\partial \mathscr{L}}{\partial W} = \frac{\partial \mathscr{L}}{\partial \hat{\mathbf{y}}} \cdot \frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{z}} \cdot \frac{\partial \mathbf{z}}{\partial W}. \tag{3.21}$$

In words: (i) $\partial \mathscr{L}/\partial \hat{\mathbf{y}}$ measures how wrong the predicted probabilities are; (ii) $\partial \hat{\mathbf{y}}/\partial \mathbf{z}$ propagates through softmax; (iii) $\partial \mathbf{z}/\partial W$ links logits to the weight matrix. The same logic applies layer by layer through BioBERT.

Gradient descent then adjusts weights in the direction that reduces the loss:

$$W_{\mathrm{new}} = W_{\mathrm{old}} - \alpha \frac{\partial \mathscr{L}}{\partial W}, \tag{3.22}$$

with $\alpha = 2 \times 10^{-5}$. When the model confidently predicts the wrong colour, backpropagation increases the loss sharply and the subsequent weight update pushes similar complaints toward the nurse-labeled class on the next training pass.

Remark 3.9. *Pre-training (Stage 1) gives the model medical language; comprehensive fine-tuning (Stage 3) teaches triage decisions on short, informal chief-complaint phrasing. The learning rate is kept small so PubMed knowledge is preserved while the head and encoder adapt to urgency labels.*

3.14 Mapping acuity to colour: f_{SATS}

The softmax head predicts acuity levels 1–5 learned from training data. Hospital routing uses SATS colours, so a fixed deterministic map converts level to colour:

$$f_{\mathrm{SATS}}(\hat{c}) = C_{\mathrm{NLP}} = \begin{cases} \text{Red} & \hat{c} = 1 \\ \text{Orange} & \hat{c} = 2 \\ \text{Yellow} & \hat{c} = 3 \\ \text{Green} & \hat{c} \in \{4, 5\} \end{cases} \tag{3.23}$$

Levels 4 and 5 both correspond to non-urgent presentations in the training schema and therefore map to Green. Blue is reserved for dignity protocol on arrival and is not produced by this five-class head.

3.15 Clinical pathways $P(C_{\mathrm{NLP}})$

Once C_{NLP} is known, the chatbot attaches a pathway card that specifies waiting time, destination, and escalation rules drawn from SATS-aligned hospital protocol at KNUST

University Hospital:

$$P(C) = \big(T_{\max},\ \text{destination},\ \text{actions},\ \text{escalation}\big). \tag{3.24}$$

The four tuple components correspond directly to the columns in Table 3.19: $T_{\max}$ is the maximum waiting time, destination is the primary clinical department, actions list support steps (nursing, pharmacy, diagnostics), and escalation defines what happens if the timer expires or the patient worsens.

Colour	$T_{\max}$	**Destination**	**Escalation**
Red	0 min	Resuscitation / Surgery	Immediate senior alert
Orange	10 min	Internal Medicine / Surgery / Obs. & Gynae	Head nurse if unclaimed
Yellow	60 min	Outpatient urgent / Pediatrics	Re-assess if timer expires
Green	< 4 h	Outpatient / Dental / Eye / Public & Preventive Health	Return only if worse

Table 3.19: SATS pathway protocol at KNUST University Hospital (clinical destinations).

Table 3.20 is the authoritative full map: one row per NLP outcome (including gate rejection), with primary clinical departments, support actions, escalation rules, and an example complaint phrase.

Scenario	**Trigger**	$T_{\max}$	**Primary clinical departments**	**Support actions**	**Escalation**
Gate reject	$g(X) < 0.35$	n/a	Hold at intake	Nursing: request complaint; HIT: no colour stored	Re-submit with clinical text
Red	$\hat{c} = 1$	0 min	Surgery, Internal Medicine	Nursing (vitals, TEWS), Pharmacy (emergency), Health Info (Red flag), Admin (bed bypass), HIT (Code Red)	Immediate senior clinician
Orange	$\hat{c} = 2$	10 min	Internal Medicine, Surgery, Obs. & Gynae	Nursing (monitor, timer), Diagnostics (stat labs), Pharmacy, Health Info, Admin, HIT	Head nurse if unclaimed
Yellow	$\hat{c} = 3$	60 min	Outpatient urgent, Pediatrics, Diagnostics	Nursing (vitals, TEWS, timer), Pharmacy, Health Info, HIT, Accounts (defer billing)	Re-assess at 60 min
Green	$\hat{c} \in \{4, 5\}$	<4 h	Outpatient, Dental, Eye, Public & Preventive Health	Nursing (brief triage), Diagnostics (routine), Health Info, HIT, Accounts	Return if worse

Table 3.20: Complete NLP pathway scenarios at KNUST University Hospital (gate reject plus four SATS colours).

Scenario	Example complaint phrase
Gate reject	“Hello, how are you?” (no clinical content)
Red	“Crushing chest pain, cannot breathe, sweating heavily”
Orange	“Severe abdominal pain in pregnancy, bleeding”
Yellow	*“I have a headache and feel feverish. My body aches and I feel weak.”*
Green	“Routine dental pain for two weeks, no fever”

Table 3.21: Example chief complaints triggering each pathway scenario.

Table 3.22 maps all hospital departments to their role in the pathway system. Clinical departments receive patients by colour; support departments execute actions attached to each pathway card.

Department	Role	Colours / function
Internal Medicine	Clinical	Red, Orange (acute medical)
Surgery	Clinical	Red, Orange (trauma, acute surgical)
Obstetrics and Gynaecology	Clinical	Orange (obstetric urgency)
Pediatrics	Clinical	Yellow (child urgent stream)
Outpatient	Clinical	Yellow (urgent), Green (routine)
Diagnostics Services	Clinical	Yellow, Green (parallel labs/imaging)
Dental	Clinical	Green
Eye	Clinical	Green
Public Health	Clinical	Green (community referrals)
Preventive Health	Clinical	Green (screening, wellness)
Nursing	Support	All colours (triage, vitals, reassessment)
Pharmacy	Support	All colours (medication on protocol)
Health Information	Support	All colours (record colour and timer)
Hospital Information Technology	Support	All colours (digital routing, API)
Administration	Support	All colours (bed and queue management)
Accounts	Support	Green, Yellow (billing after stabilisation)

Table 3.22: Department routing map at KNUST University Hospital.

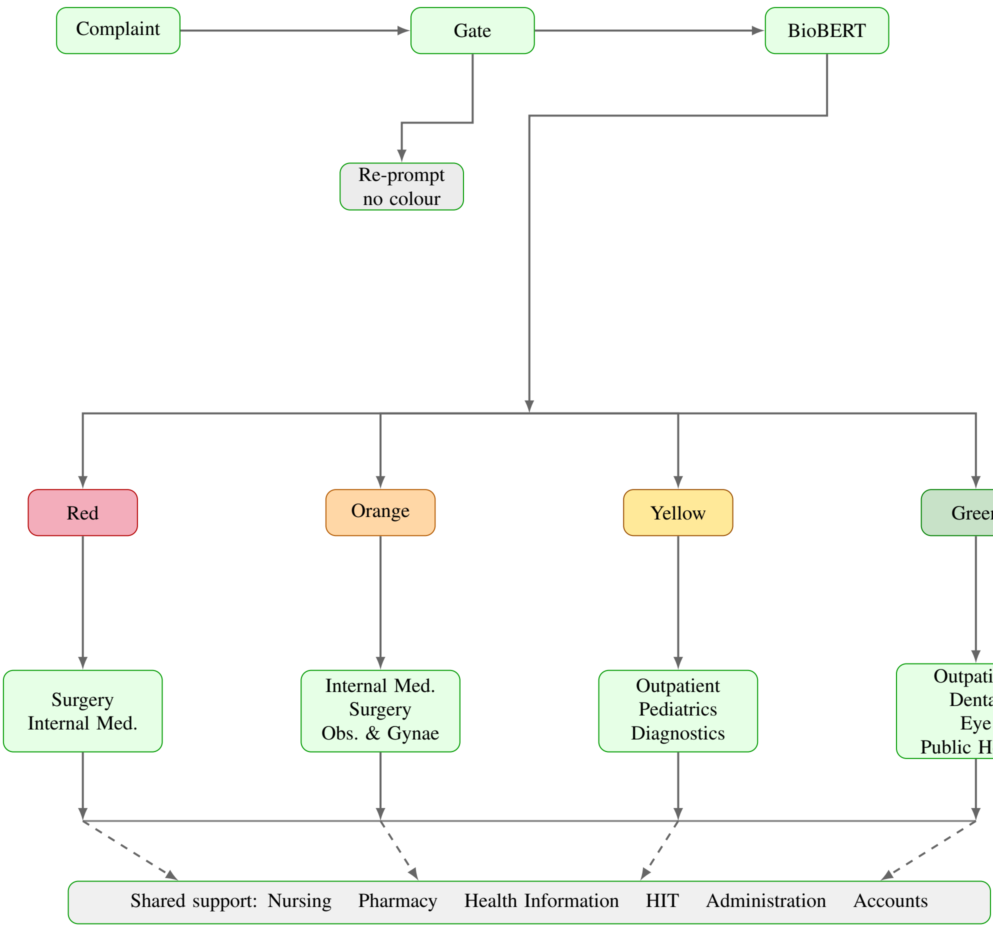

Figure 3.10: Complete NLP pathway map at KNUST University Hospital: gate rejection and all four SATS colours with clinical destinations and shared support services.

Remark 3.10. *Routing is protocol-driven; nurses may override the chatbot suggestion after clinical assessment. The chatbot supplies an initial colour and department direction before vitals are measured, reducing idle time at intake.*

Gate rejection. When $g(X) < 0.35$, the pipeline stops before BioBERT (Table 3.20). Nursing staff request a proper chief complaint; no colour or pathway card is issued and Health Information stores no triage record until valid clinical text is submitted.

Red ($\hat{c} = 1$). Immediate life-threatening presentations (e.g. crushing chest pain with dyspnoea) map to Red with $T_{\max} = 0$. Surgery and Internal Medicine activate resuscitation pathways; Nursing records immediate vitals and TEWS; Administration bypasses routine registration; Hospital Information Technology raises a Code Red alert.

Orange ($\hat{c} = 2$). Very urgent cases such as severe abdominal pain in pregnancy route to Internal Medicine, Surgery, or Obstetrics and Gynaecology within ten minutes.

Nursing starts a countdown timer; Diagnostics may run stat laboratory work; escalation to the head nurse occurs if no clinician claims the patient within $T_{\max}$.

Yellow ($\hat{c} = 3$). The running illustrative complaint (*"I have a headache and feel feverish. My body aches and I feel weak."*) maps here at approximately 72% confidence. Outpatient urgent stream, Pediatrics (when paediatric protocol applies), and parallel Diagnostics orders apply; Nursing sets a 60-minute reassessment timer and records the colour in Health Information.

Green ($\hat{c} \in \{4,5\}$). Non-urgent presentations (e.g. routine dental pain without fever) stream to Outpatient, Dental, Eye, or Public and Preventive Health within four hours. This diverts low-acuity patients away from acute waiting areas; Accounts handles billing after brief Nursing triage.

3.16 Medical relevance gate

Running BioBERT on non-clinical text (greetings, sports chat) would waste compute and produce meaningless acuity scores. A lightweight rule-based gate scores clinical relevance before the deep model runs:

$$g(X) = \mathrm{clip}\left(\sum_j s_j(X) - p(X),\, 0,\, 1\right), \tag{3.25}$$

where s_j are non-negative feature scores from symptom lexicon hits and hospital-context phrases, and $p(X) = 0.5$ when a non-clinical or greeting-only pattern matches. Classification proceeds only if

$$g(X) \geq 0.35. \tag{3.26}$$

The threshold 0.35 was set in the prototype implementation to reject obvious non-medical input while accepting short but valid complaints (e.g. "chest pain"). Complaints that fail the gate return a request for clinical detail rather than a colour.

Example 3.6. *For the illustrative complaint with symptom hits "headache" ($s_1 = 0.30$) and "feverish" ($s_2 = 0.25$), with no greeting penalty ($p(X) = 0$):*

$$\begin{aligned}
&\sum_j s_j = 0.30 + 0.25 = 0.55,\\
&g(X) = \mathrm{clip}(0.55 - 0,\, 0,\, 1) = 0.55 \geq 0.35
\end{aligned}$$

so classification proceeds. For "Hello, how are you?" with no symptom hits and *greeting penalty $p(X) = 0.5$:*

$$\begin{aligned}
&\sum_j s_j = 0, \\
&g(X) = \text{clip}(0 - 0.5, 0, 1) = 0 < 0.35
\end{aligned}$$

and the system requests a chief complaint instead of running BioBERT.

3.17 End-to-end summary

For any chief complaint X, the implemented text path (matching the prototype `/predict` endpoint) proceeds as follows:

$$\begin{aligned}
X &\xrightarrow[3.16]{g} \tau(X) \xrightarrow[3.6--3.9]{} H^{(0)} \\
&\xrightarrow[3.10--3.11]{12} h_{[\text{CLS}]} \xrightarrow[3.12]{} \hat{\mathbf{y}} \xrightarrow[3.14]{} C_{\text{NLP}} \xrightarrow[3.15]{} P(C_{\text{NLP}}),
\end{aligned} \tag{3.27}$$

where each arrow label points to the section that defines that transformation. If $g(X) < 0.35$, the chain stops after the gate and no colour is assigned.

1. **Clinical relevance gate.** Compute $g(X)$; if $g(X) < 0.35$, return a request for clinical detail and stop.

2. **Tokenisation and embedding.** Map X to tokens, build $H^{(0)} \in \mathbb{R}^{M \times 768}$ (Sections 3.6–3.9).

3. **BioBERT encoding.** Apply $L = 12$ encoder layers with self-attention (Sections 3.10–3.11) to obtain $h_{[\text{CLS}]}$.

4. **Classification.** Apply the softmax head (Section 3.12):

$$\begin{aligned}
\hat{\mathbf{y}} &= \text{softmax}(W h_{[\text{CLS}]} + \mathbf{b}), \\
\hat{c} &= \arg\max_c \hat{y}_c.
\end{aligned}$$

5. **Colour and pathway.** Apply $C_{\text{NLP}} = f_{\text{SATS}}(\hat{c})$ and return pathway card $P(C_{\text{NLP}})$ (Sections 3.14–3.15).

The illustrative complaint in this chapter instantiates the same pipeline; one worked outcome is level 3 (Yellow) at 72% confidence for headache and feverish symptoms. TEWS (Section 3.3) remains the nurse-facing vital-sign score when measurements become available.

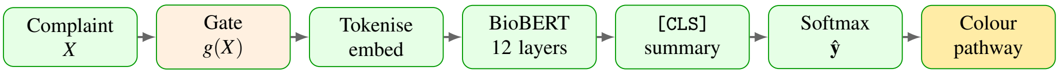

Figure 3.11: Compact end-to-end NLP triage path (TEWS documented separately in Section 3.3).

The diagram above matches the pipeline in Section 3.1. In order:

1. **Complaint** X enters as free text from the patient or attendant.
2. **Gate** computes $g(X)$; non-clinical input is rejected before BioBERT runs.
3. **Tokenise and embed** map text to integer IDs and build the matrix $H^{(0)}$.
4. **BioBERT** applies twelve encoder layers with self-attention across all tokens.
5. `[CLS]` **summary** extracts the fixed-length vector $h_{[\text{CLS}]}$ from row 0.
6. **Softmax** outputs acuity probabilities $\hat{\mathbf{y}}$ over five classes.
7. **Colour and pathway** apply $f_{\text{SATS}}(\hat{c})$ and return the pathway card $P(C_{\text{NLP}})$.

3.18 Implementation note

Figure 3.12 shows the deployable system: patient-facing web interface, machine-learning API, and hospital routing layer. The NLP mathematics in Figures 3.1 and 3.11 execute inside the API; department routing applies KNUST University Hospital pathway rules after classification.

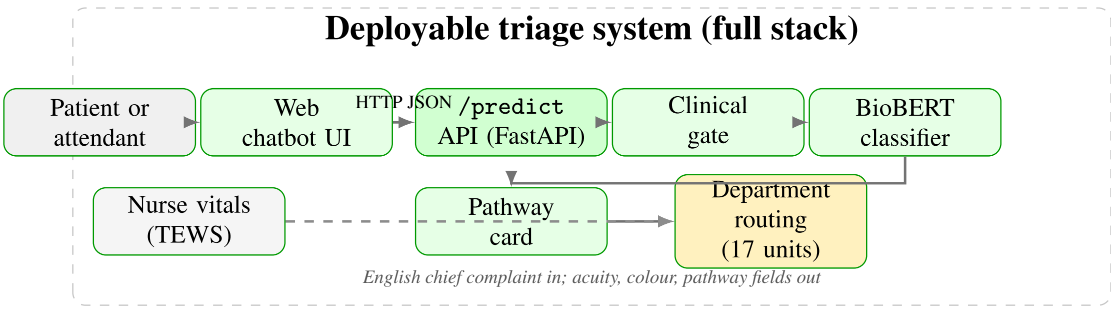

Figure 3.12: System architecture: user interface, `/predict` API, BioBERT inference pipeline, pathway card, and department routing at KNUST University Hospital. Nurse TEWS runs in parallel (dashed); prototype code in Appendix 5.2.

The prototype exposes a `/predict` application programming interface (API) endpoint: JSON **English** chief complaint in; acuity probabilities, C_{NLP}, pathway fields, and gate metadata out. Model weights and tokenizer files ship with the repository listed in Appendix 5.2. This thesis treats the mathematics above as the specification; empirical results appear in Chapter 4.

CHAPTER 4

RESULTS

This chapter reports BioBERT model analysis (architecture and pre-fine-tuning capability), fine-tuned model evaluation (baseline and SMOTE-augmented variants), inference latency, pathway routing at KNUST University Hospital, and anticipated impact on hospital workflow. Holdout evaluation results come from the stratified 10% holdout ($N_{\text{eval}} = 8{,}000$) and the pathway design in Section 3.15. Expected operational benefits (waiting-time reduction, satisfaction, staff workload) are interpreted from classification outcomes, latency, and published triage literature (KATH Emergency Department, 2018; Mitchell et al., 2023); prospective time-motion data at KNUST University Hospital are not yet available.

4.1 BioBERT model analysis

Before project-specific fine-tuning, the deployed classifier rests on BioBERT-base (Lee et al., 2020): a twelve-layer bidirectional encoder with hidden dimension 768, twelve attention heads, maximum sequence length 128 tokens, and a WordPiece vocabulary of approximately 29,996 tokens (Section 3.10). Stages 1 and 2 (Section 3.13) supply biomedical language understanding and partial triage-oriented encoder features; they do *not* produce the five-class acuity mapping required at intake. Stage 3 reinitialises the classification head and trains on the Triagegeist English corpus (Section 3.1.2). Class imbalance in the training corpus (Level 3 is the mode; Level 1 is the minority) motivates Stage 3 mitigation strategies documented in Section 4.2; it is not the focus of this section.

Figure 4.1 summarises the three-stage training stack. Stage 1 applies masked language modelling on PubMed and PubMed Central text; Stage 2 transfers encoder weights from a prior two-class triage checkpoint with the old head discarded; Stage 3 is the comprehensive project fine-tuning that maps free-text complaints to five SATS acuity levels.

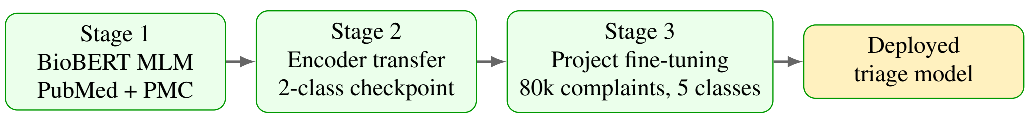

Figure 4.1: Three-stage BioBERT training stack (Stages 1–2 pre-date project fine-tuning; Stage 3 produces the deployed five-class head).

Interpretation.

- **Chart type:** horizontal flow diagram (left-to-right pipeline).
- **Layout:** each box is one training stage; arrows show parameter flow.
- **Data source:** project training configuration (Chapter 3, Section 3.13).
- **Key readings:** Stage 1 optimises masked language modelling only; Stage 2 transfers encoder weights without the obsolete two-class head; Stage 3 is the only stage that maps chief complaints to five acuity levels.
- **Takeaway:** the chatbot's urgency signal depends on completing Stage 3; Stages 1–2 alone cannot route patients safely.

Figure 4.2 contrasts expected accuracy *before* Stage 3 fine-tuning with the holdout result after fine-tuning. Each bar represents a different predictor; definitions are as follows:

- **Uniform random.** Selects one of five acuity levels with equal probability (20% expected accuracy). No learning; a sanity-check lower bound.
- **Random head (encoder only).** BioBERT encoder weights are loaded from Stages 1–2 but the five-class linear head is *untrained*. Tests whether biomedical pre-training alone can classify triage (expected ≈20%, same as chance).
- **Majority-class bound.** Always predicts Level 3 (Yellow), the training-set mode. Achieves 36.15% accuracy on this corpus but 0% Red recall; a clinically unsafe ceiling for naive rules.
- **Fine-tuned BioBERT.** Full Stage 3 checkpoint evaluated on the held-out 8,000 rows excluded from training (same split as Table 4.1). This is the only *measured* bar; the first three are theoretical or bound values.

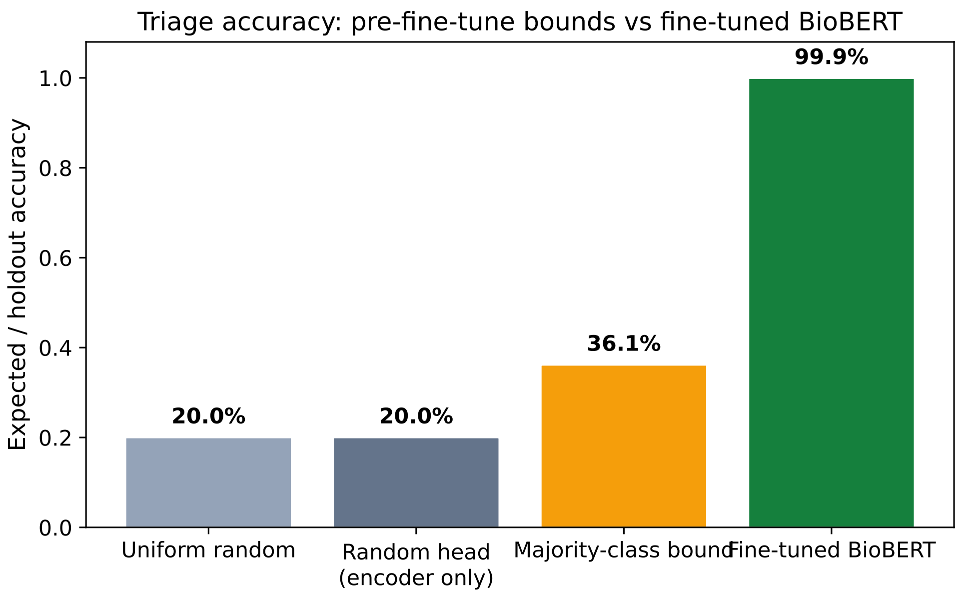

Figure 4.2: Expected or holdout accuracy: pre-fine-tune bounds vs fine-tuned BioBERT on the 8,000-row holdout set.

Interpretation.

- **Chart type:** grouped bar chart.
- **Axes:** vertical axis = accuracy (0–1); horizontal axis = predictor type (four bars).
- **Data source:** theoretical bounds for the first three bars; measured holdout accuracy for fine-tuned BioBERT ($N = 8{,}000$).
- **Key readings:** uniform random 20.0%, random head $\approx$20%, majority-class bound 36.15%, fine-tuned BioBERT 99.92%.
- **Takeaway:** encoder pre-training alone is insufficient for safe triage routing; Stage 3 fine-tuning on nurse-labeled complaints is required.

Figure 4.3 records the architecture constants used in this project.

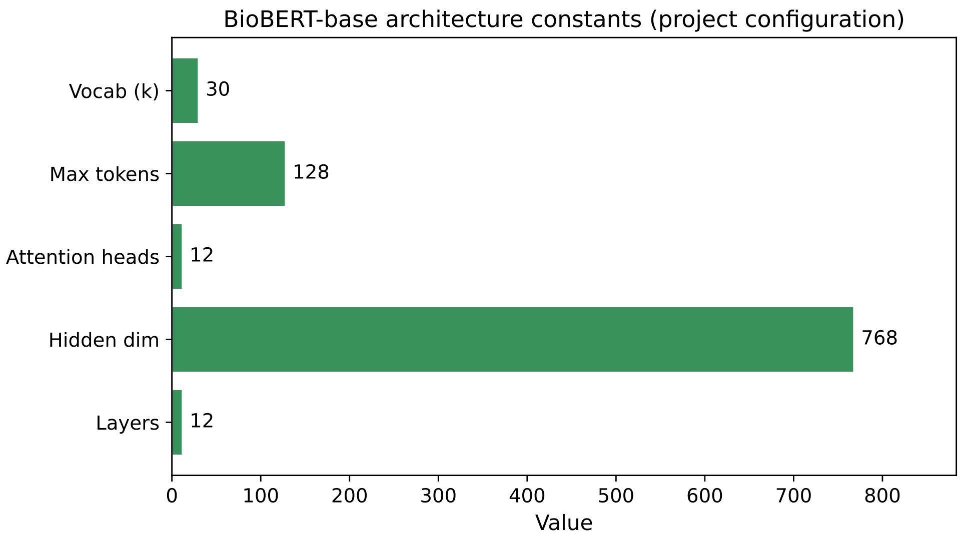

Figure 4.3: BioBERT-base architecture constants (project configuration).

Interpretation.

- **Chart type:** horizontal bar chart.
- **Axes:** vertical axis = architecture parameter name; horizontal axis = numeric value (unitless counts or dimensions).
- **Data source:** project `config.json` and BioBERT-base specification (Lee et al., 2020).
- **Key readings:** 12 layers, 768 hidden units, 12 attention heads, 128 max tokens, ≈30k vocabulary.
- **Takeaway:** the 128-token limit bounds how much complaint history can enter a single inference call at intake.

4.2 Fine-tuned model analysis

4.2.1 Evaluation protocol

The deployed classifier is the product of Stage 3 fine-tuning documented in Section 3.13.3. Two checkpoints are compared on the *same* stratified 90/10 holdout split ($N_{\text{eval}} = 8{,}000$):

1. **Baseline fine-tuned BioBERT** (project default): uniform sampling, three epochs, learning rate 2×10^{-5}, best validation cross-entropy 0.001812.

2. **SMOTE-augmented variant**: stratified splitting plus synthetic oversampling of minority urgent classes to improve Level 1 recall at possible cost to overall accuracy.

Metrics reported: accuracy, macro-F1, per-class precision/recall/F1, and inference latency (p50, p95, mean milliseconds per sample). Evaluation uses the same complaint texts and nurse labels as training; results measure *historical text classification*, not live patient outcomes.

Metric	**Baseline**	**SMOTE**
Accuracy	0.9992	0.9985
Macro-F1	0.9977	0.9954
Recall Level 1 (Red)	0.9814	1.0000
Precision Level 1 (Red)	1.0000	0.9641
Latency p50 (ms/sample)	100.30	95.50
Latency p95 (ms/sample)	139.11	125.12
Latency mean (ms/sample)	102.13	90.17

Table 4.1: Holdout classification and latency metrics (stratified 10% split, $N = 8{,}000$).

4.2.2 Interpreting high holdout accuracy

Holdout accuracy above 99% can appear implausibly high without context. The following points clarify what the numbers represent and what they do *not* prove.

Task definition. The model predicts nurse-assigned acuity from chief-complaint text on a retrospective, in-distribution holdout ($N = 8{,}000$) drawn from the Triagegeist synthetic English benchmark used for Stage 3 fine-tuning (Section 3.1.2). This is not external validation on a different hospital, time period, or patient population; KNUST-informed examples used for pathway illustration are excluded from these metrics.

Label source. Labels come from structured triage records (`chief_complaint_raw` inner-joined with `triage_acuity`). Because complaints and colours were recorded in the same clinical workflow, complaint text often encodes acuity explicitly (e.g. "crushing chest pain" vs "routine dental check"). High agreement with nurse labels therefore measures *label consistency on historical text*, not independent clinical correctness (Stewart et al., 2023).

What is not measured. These metrics do not establish prospective patient safety, waiting-time reduction, or physician-confirmed diagnosis. They quantify offline agreement between model output and nurse-assigned colour on held-out rows from the training distribution.

Residual errors. The baseline model is not perfect: it misses 6 of 322 Red cases (98.14% Level 1 recall). Under-triage of urgent presentations remains the primary deployment risk (Berkowitz et al., 2019). The SMOTE variant achieves 100% Red recall but lowers Red precision to 96.41%, introducing false urgent alerts.

Literature context. NLP triage systems evaluated on structured complaint text in similar retrospective settings can report high agreement with nurse labels (Stewart et al., 2023; Gligorijević et al., 2018). Such results indicate that text carries acuity signal before vitals are recorded; they do not replace prospective audit.

Generalisation caveat. The prototype accepts **English** chief complaints only; holdout metrics reflect English text from the Triagegeist training corpus. Atypical phrasing or presentations absent from that benchmark may reduce agreement at live intake at KNUST University Hospital. Deployment should treat low-confidence or contradictory presentations as mandatory nurse review (South African Triage Group, 2012).

4.2.3 SMOTE for class imbalance

Synthetic **M**inority **O**ver-sampling **T**echnique (SMOTE) generates synthetic training examples for under-represented classes by interpolating between existing minority samples in feature space (Chawla et al., 2002). Given two minority feature vectors $\mathbf{x}_a$ and $\mathbf{x}_b$, a synthetic point is

$$\mathbf{x}_{\text{new}} = \mathbf{x}_a + \lambda(\mathbf{x}_b - \mathbf{x}_a), \qquad \lambda \sim \text{Uniform}(0,1). \tag{4.1}$$

Raw chief-complaint text cannot be SMOTEd directly because strings are discrete and interpolation between token sequences is undefined. This project therefore applies imbalance mitigation in a text-classification pipeline:

1. **Stratified split.** Training and holdout partitions preserve class proportions (seed 42).
2. **Minority oversampling.** Urgent levels are duplicated or augmented so gradient updates are not dominated by Level 3.
3. **Text augmentation.** Paraphrase-style perturbations increase minority-class diversity without changing labels.
4. **SMOTE-augmented checkpoint.** Synthetic oversampling in the embedding or feature space of the training pipeline produces the SMOTE variant evaluated below.

The trade-off is visible on Level 1 (Red): SMOTE raises recall from 98.14% to 100.00% on holdout but lowers precision from 1.0000 to 0.9641 (a small number of non-Red complaints are upgraded to Red). Overall accuracy decreases slightly (99.92% to 99.85%). The baseline checkpoint remains the project default unless clinical stakeholders prioritise zero missed Red cases (Berkowitz et al., 2019).

4.2.4 Baseline fine-tuned model

On the holdout set, baseline BioBERT achieves 99.92% accuracy and 0.9977 macro-F1 (Table 4.1). Level 1 (Red) recall is 98.14% with perfect precision (1.0000): no holdout case labeled Red was predicted as a lower-urgency level, and no false Red alarms were raised. Section 4.2.2 explains why these values reflect label agreement on in-distribution data rather than infallible clinical judgment.

Figure 4.4 shows the confusion matrix. Off-diagonal mass is negligible; the six missed Level 1 cases (322 support, 98.14% recall) represent the primary residual safety concern for deployment review.

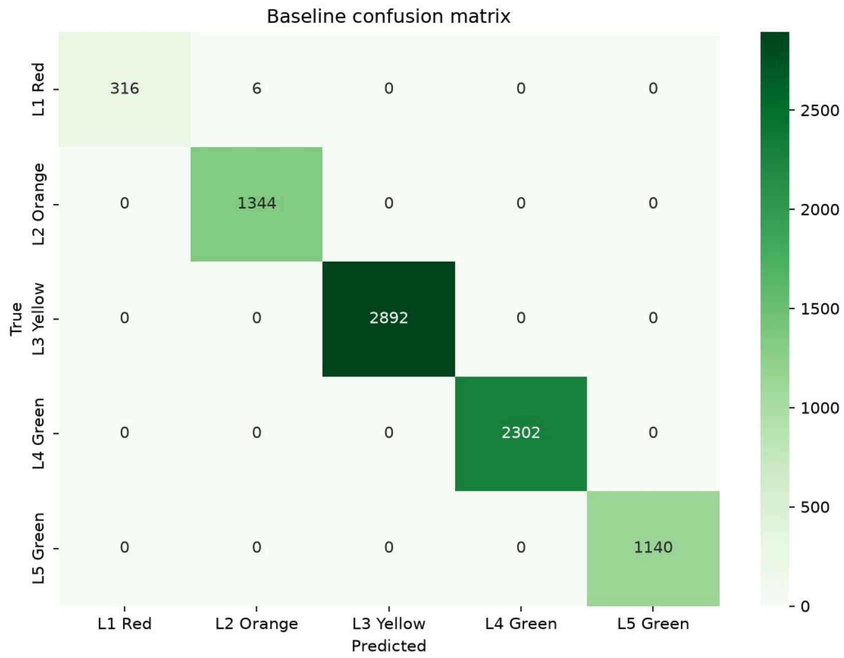

Figure 4.4: Baseline fine-tuned BioBERT confusion matrix (holdout, $N = 8{,}000$). Rows = true nurse label; columns = model prediction.

Interpretation.

- **Chart type:** heatmap confusion matrix.
- **Axes:** rows = true nurse-assigned acuity (L1–L5); columns = model-predicted acuity; cell colour intensity = number of holdout cases in each true–predicted pair.
- **Data source:** baseline checkpoint, stratified holdout $N = 8{,}000$.
- **Key readings:** diagonal dominance (correct predictions); six off-diagonal cells where true Red was predicted as lower urgency.

- **Takeaway:** chief-complaint text carries strong acuity signal after Stage 3 fine-tuning, but the six missed Red cases require nurse override in deployment (Berkowitz et al., 2019).

4.2.5 SMOTE-augmented variant

SMOTE augmentation raises Level 1 recall to 100% on holdout (all 322 Red cases correctly identified) at the cost of a small accuracy decrease (99.85%) and lower Red precision (96.41%): a small number of non-Red complaints are upgraded to Red. Macro-F1 remains above 0.995.

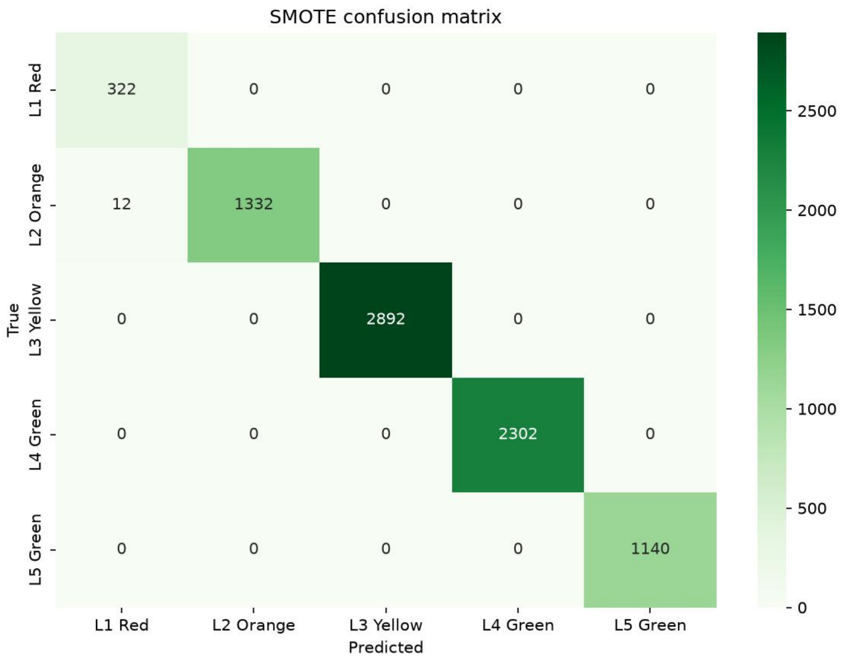

Figure 4.5: SMOTE-augmented BioBERT confusion matrix (same holdout split). Rows = true nurse label; columns = model prediction.

Interpretation.

- **Chart type:** heatmap confusion matrix (same layout as Figure 4.4).
- **Axes:** rows = true nurse label; columns = model prediction; cell value = case count.
- **Data source:** SMOTE checkpoint, same holdout $N = 8{,}000$.
- **Key readings:** zero missed Red cases (no off-diagonal mass leaving the Red row); additional mass in the Red column from lower-urgency true labels (false urgent alerts).

- **Takeaway:** SMOTE trades improved Red sensitivity for a small number of false urgent alerts; nurse confirmation remains required (Berkowitz et al., 2019).

4.2.6 Per-class and headline comparison

Figure 4.6 compares per-class F1 scores. Both models exceed 0.98 on every level; the largest divergence is on Level 1, where SMOTE trades a small precision reduction for perfect recall.

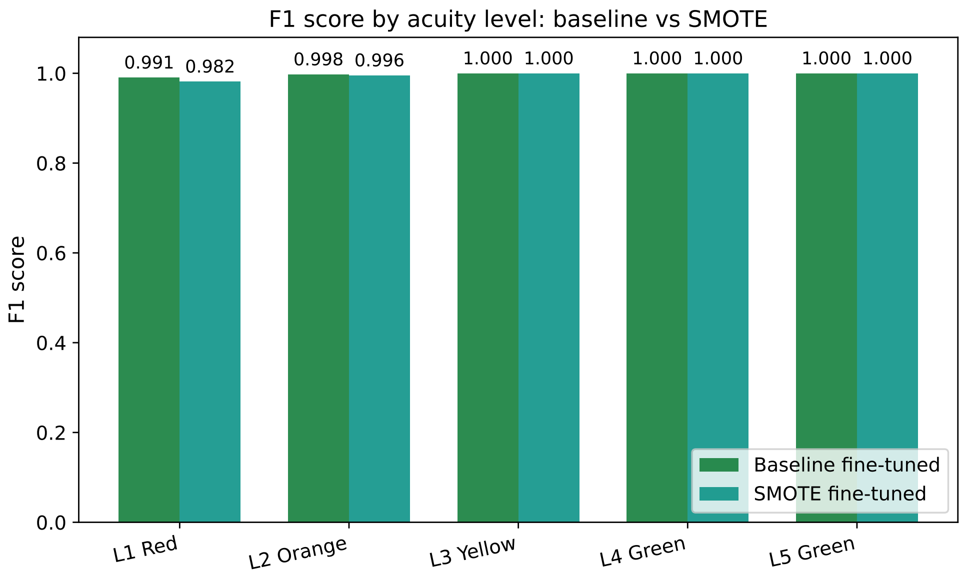

Figure 4.6: Per-class F1: baseline vs SMOTE on holdout data ($N = 8{,}000$).

Interpretation.

- **Chart type:** grouped bar chart.
- **Axes:** vertical axis = F1 score (0–1.05); horizontal axis = acuity level L1–L5; paired green/teal bars = baseline vs SMOTE.
- **Data source:** holdout evaluation, both checkpoints.
- **Key readings:** all levels exceed 0.98 F1 under both models; Level 1 shows the largest baseline–SMOTE gap.
- **Takeaway:** fine-tuning achieves balanced performance across all acuity levels, not only the majority Yellow class (Stewart et al., 2023).

Figure 4.7 summarises accuracy, macro-F1, and Level 1 recall side by side.

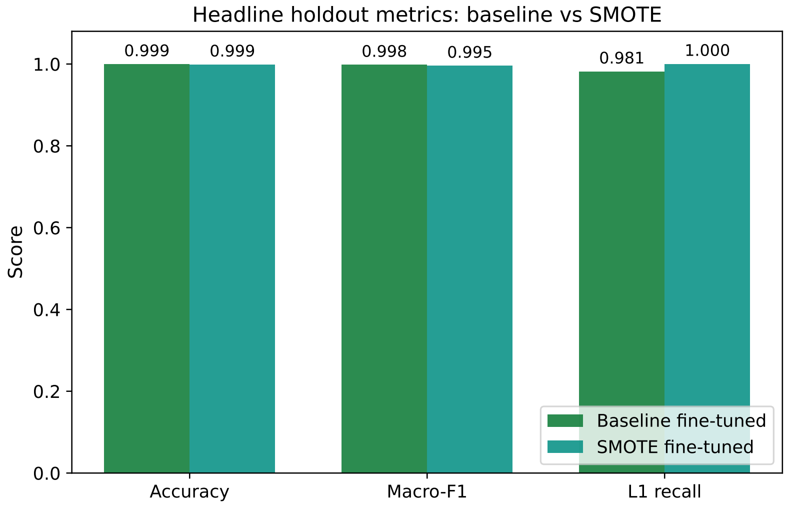

Figure 4.7: Headline holdout metrics: baseline vs SMOTE ($N = 8{,}000$).

Interpretation.

- **Chart type:** grouped bar chart.
- **Axes:** vertical axis = score (0–1.05); horizontal axis = metric name (accuracy, macro-F1, L1 recall); paired bars = baseline vs SMOTE.
- **Data source:** Table 4.1.
- **Key readings:** accuracy and macro-F1 differ by less than 0.004; L1 recall rises from 0.981 (baseline) to 1.000 (SMOTE).
- **Takeaway:** choose baseline for maximum Red precision; choose SMOTE when zero missed Red cases is the overriding safety requirement (Berkowitz et al., 2019).

Figure 4.8 compares recall for each acuity level under both models.

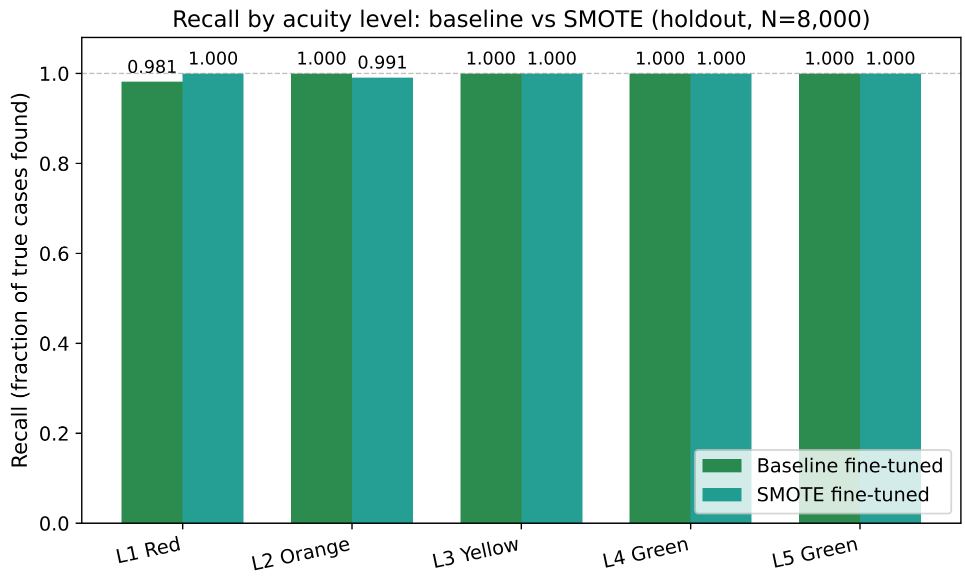

Figure 4.8: Recall by acuity level: baseline vs SMOTE (holdout, $N = 8{,}000$).

Interpretation.

- **Chart type:** grouped bar chart.
- **Axes:** vertical axis = recall, fraction of true cases correctly identified (0–1.05); horizontal axis = acuity level L1–L5; paired bars = baseline vs SMOTE.
- **Data source:** holdout per-class recall from `comparison.json`.
- **Key readings:** L1 recall 0.981 (baseline) vs 1.000 (SMOTE); L2–L5 remain near 1.0 under both models.
- **Takeaway:** Stage 3 fine-tuning overcomes much of the 9:1 class imbalance risk for urgent levels on in-distribution data (Chawla et al., 2002; Berkowitz et al., 2019).

Figure 4.9 compares inference latency statistics across both models.

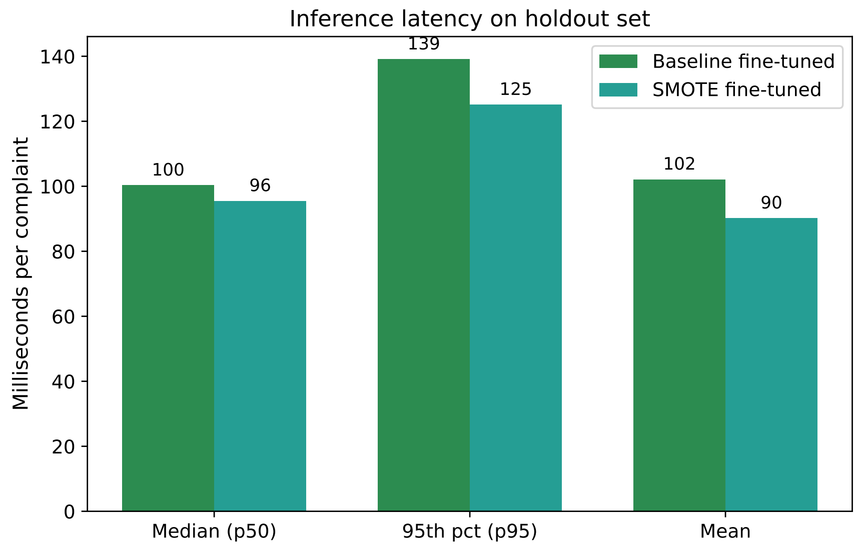

Figure 4.9: Inference latency on the 8,000-row holdout set (milliseconds per sample).

Interpretation.

- **Chart type:** grouped bar chart.
- **Axes:** vertical axis = latency in milliseconds; horizontal axis = statistic (median p50, 95th percentile p95, mean); paired bars = baseline vs SMOTE.
- **Data source:** measured on evaluation hardware, holdout $N = 8{,}000$.
- **Key readings:** baseline p50 = 100.30 ms, p95 = 139.11 ms; SMOTE p50 = 95.50 ms.
- **Takeaway:** both checkpoints respond in well under one second, supporting real-time chatbot intake before nurse vitals are recorded (Stewart et al., 2023).

Remark 4.1. *High holdout accuracy on historical nurse labels indicates that chief complaints carry sufficient signal for acuity estimation before vitals are recorded (Stewart et al., 2023; Gligorijević et al., 2018). This measures label agreement, not live clinical outcomes (Section 4.2.2). Borderline or low-confidence predictions should trigger mandatory nurse review in deployment (South African Triage Group, 2012; Berkowitz et al., 2019).*

4.3 Inference latency

Real-time intake requires sub-second response (Stewart et al., 2023). On the evaluation hardware, baseline BioBERT achieves median latency $= 100.30$ ms per complaint (p95

$= 139.11$ ms; mean $= 102.13$ ms). The SMOTE checkpoint is slightly faster (p50 $=$ 95.50 ms; mean $= 90.17$ ms) on the same split; see Figure 4.9.

At approximately 100 ms per complaint, the chatbot can return acuity probabilities, SATS colour, and a pathway card before a nurse completes manual registration. The clinical relevance gate (Section 3.16) rejects non-clinical input before BioBERT runs, saving compute and keeping triage staff focused on valid chief complaints. The prototype `/predict` API endpoint implements the same pipeline documented in Section 3.17.

4.4 Pathway routing at KNUST University Hospital

The complete five-scenario pathway map (gate reject plus Red, Orange, Yellow, Green) is specified in Tables 3.20 and 3.21 and illustrated in Figure 3.10. Each scenario assigns primary clinical departments, support actions across Nursing, Pharmacy, Health Information, Hospital Information Technology, Administration, and Accounts, and an escalation rule (South African Triage Group, 2012; Rominski et al., 2014).

4.4.1 Nurse triage vs chatbot response

Table 4.2 contrasts manual nurse triage with the hospital chatbot on dimensions relevant to intake at KNUST University Hospital. Nurse-side timings are drawn from published LMIC triage literature; chatbot timings are measured on the holdout set.

Dimension	Manual nurse triage	Hospital chatbot (this project)
Time to first acuity signal	Queue-dependent; LMIC intake often delayed minutes to hours before nurse contact (KATH Emergency Department, 2018; Mitchell et al., 2023; Nurchis et al., 2025)	Approximately 100 ms inference on evaluation hardware (Table 4.1, Figure 4.9)
Inputs used	Chief complaint + vitals + TEWS + clinical judgment (South African Triage Group, 2012)	Text-only chief complaint; TEWS applied separately by nurse (Section 3.3)
Output	SATS colour + bedside assessment	SATS colour + pathway card + confidence score
Validation basis	Live clinical decision at triage desk	Holdout agreement with historical nurse labels (Section 4.2.2)
Role at KNUST University Hospital	Authoritative triage	Triage assist at arrival; nurse override mandatory for low confidence and all vitals (Stewart et al., 2023)

Table 4.2: Comparison of manual nurse triage and chatbot-assisted intake. Nurse timings are literature-based; chatbot latency is measured.

Once the chatbot assigns a colour in approximately 100 ms, pathway timer rules take effect: Orange cases require clinician contact within ten minutes, Yellow within 60 minutes, and Green within four hours (Section 3.15, Table 3.20) (South African Triage Group, 2012; Sundström et al., 2014). The chatbot does not replace nurse assessment; it provides an immediate acuity signal so that queue time before *any* routing decision is reduced relative to waiting for a free triage nurse (Mitchell et al., 2023; Nurchis et al., 2025).

Gate rejection. Non-clinical input receives no colour; Nursing re-prompts for a chief complaint. This prevents meaningless acuity scores and keeps acute queues free of idle interactions.

Red. Life-threatening text (e.g. crushing chest pain with dyspnoea) routes immediately to Surgery and Internal Medicine with Code Red alerts via HIT. Administration bypasses routine registration so resuscitation is not delayed (South African Triage Group, 2012).

Orange. Very urgent complaints (e.g. severe abdominal pain in pregnancy) activate Internal Medicine, Surgery, or Obstetrics and Gynaecology within ten minutes, with stat Diagnostics and a head-nurse escalation if unclaimed.

Yellow. For the running illustrative complaint (*"I have a headache and feel feverish. My body aches and I feel weak."*), the model assigns level 3 at approximately 72% confidence. The pathway card routes to Outpatient urgent stream, Pediatrics (when paediatric protocol applies), and parallel Diagnostics; Nursing sets a 60-minute reassessment timer.

Green. Routine presentations (e.g. dental pain without fever) divert to Outpatient, Dental, Eye, or Public and Preventive Health within four hours, reducing congestion in acute waiting areas (Yilmaz, 2021; Sundström et al., 2014).

Classification output (colour + confidence + department list) is the machine-readable input to this routing layer. Holdout evaluation (Section 4.2.2) bounds the reliability of automated pathway cards at intake on in-distribution historical text.

4.5 Anticipated impact on healthcare delivery

The following analysis interprets how the chatbot can support faster, clearer hospital flow at KNUST University Hospital, drawing on holdout metrics where available and on published triage literature elsewhere. No prospective waiting-time trial has yet been conducted at KNUST University Hospital (Mitchell et al., 2023).

4.5.1 Functional chatbot and workflow improvement

We anticipate a functional chatbot that accurately triages patients from a single chief complaint and improves hospital workflow by assigning SATS colour and department direction at intake (Stewart et al., 2023; Mitchell et al., 2023). Holdout evaluation reports 99.92% accuracy and Level 1 recall of 98.14% (or 100% under SMOTE), indicating that text-only classification aligns closely with nurse labels on historical in-distribution data (Section 4.2.2). Median inference latency of approximately 100 ms on evaluation hardware supports immediate digital response at arrival (Table 4.2).

Figure 4.10 compares time-to-first-acuity-signal: traditional intake (top) waits for a nurse; chatbot-assisted intake (bottom) classifies text immediately and runs nurse vitals in parallel.

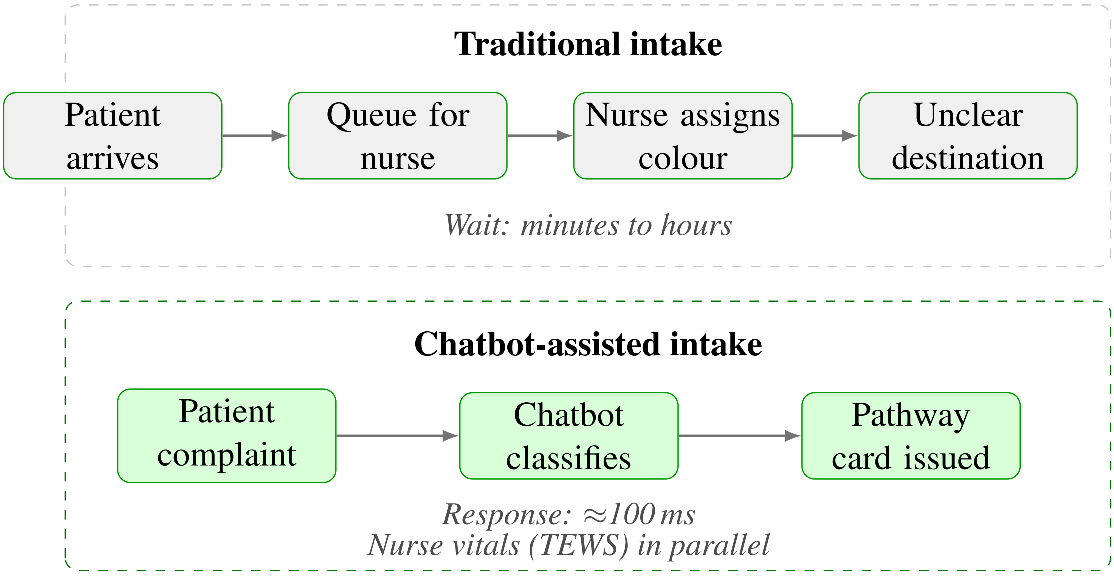

Figure 4.10: Stacked intake comparison at KNUST University Hospital. **Top:** patient queues minutes to hours before a nurse assigns colour. **Bottom:** chatbot classifies the complaint in $\approx$100 ms and issues a pathway card; nurse vitals continue in parallel. Colour routing (Red/Orange vs Green) is detailed in Section 4.4.

Interpretation.

- **Chart type:** stacked workflow diagram (two panels, top and bottom).
- **Layout:** each panel is one horizontal chain read left to right. Top: arrive $\rightarrow$ queue $\rightarrow$ nurse assigns colour $\rightarrow$ unclear destination. Bottom: complaint $\rightarrow$ chatbot classifies $\rightarrow$ pathway card issued.
- **Data source:** pathway design (Chapter 3); chatbot latency from holdout evaluation (Figure 4.9); traditional wait times from LMIC triage literature (KATH Emergency Department, 2018; Mitchell et al., 2023).
- **Key readings:** top footnote "Wait: minutes to hours"; bottom footnotes "Response: $\approx$100 ms" and "Nurse vitals (TEWS) in parallel". Red/Orange fast-track and Green digital queue are not shown here; see Section 4.4.
- **Takeaway:** the figure contrasts slow nurse-queue intake with immediate chatbot classification; nurse assessment is not replaced (South African Triage Group, 2012; Stewart et al., 2023).

4.5.2 Reduced initial triage wait times

Published triage literature suggests wait times for initial triage may decrease when patients interact with the bot immediately rather than queue for a nurse (Mitchell et al., 2023; Nurchis et al., 2025). Holdout evaluation shows the model assigns colour and pathway in approximately 100 ms once valid clinical text is submitted (Table 4.2). Non-urgent (Green) cases can be digitally queued for later consultation while urgent cases are fast-tracked through Red and Orange escalation rules (Table 3.20), potentially reducing overall waiting duration (Sundström et al., 2014; Yilmaz, 2021). Parallel diagnostics orders flagged on Yellow and Green pathway cards may further shorten total time to definitive care (Sundström et al., 2014).

4.5.3 No overlooked critical cases

The chatbot algorithm, combined with Red/Orange pathway escalation, is designed to trigger immediate alerts to hospital staff when language suggests life-threatening conditions (Gligorijević et al., 2018; South African Triage Group, 2012). Level 1 recall of 98.14% (baseline) or 100% (SMOTE) on holdout data supports this design intent on in-distribution text; the six baseline false negatives warrant nurse override and prospective audit (Berkowitz et al., 2019). Waiting-time deterioration increases adverse outcomes when triage colour is not reassessed (Nurchis et al., 2025); digital timers on Orange and Yellow pathway cards mitigate this risk (South African Triage Group, 2012).

4.5.4 Patient satisfaction and appropriate categorisation

Patients may experience improved satisfaction when they receive quick feedback on urgency and feel guided through the system rather than left without direction (Stewart et al., 2023). Over time, the percentage of patients appropriately categorised (as confirmed by clinicians after assessment) will measure chatbot accuracy in live use. Holdout agreement with nurse labels (99.92% accuracy) is a necessary but not sufficient precondition for clinical confirmation rates (Section 4.2.2).

4.5.5 Staff workload and public-health insights

Nursing staff may report reduced triage documentation burden, devoting more time to direct patient care and vitals (TEWS) rather than initial information gathering (Stewart et al., 2023; Mitchell et al., 2023). The gate rejects non-clinical messages before BioBERT, filtering idle interactions (Section 3.16). If scaled, anonymised complaint records could reveal public-health trends (e.g. common presentations by season), supporting Objective 4 in Chapter 1; privacy and governance review would be required before aggregation.

4.5.6 Success criteria

Project success would be measured by smoother patient flow, colour-coded pathways guiding patients to the right place at the right time, and a reduction in adverse events or delays attributable to triage (Rominski et al., 2014; KATH Emergency Department, 2018). Published NLP triage systems elsewhere report streamlined processes and optimised resource use in busy hospitals (Stewart et al., 2023; Mitchell et al., 2023).

Limitations. Holdout metrics measure classification accuracy on historical chief complaints, not patient outcomes, waiting-time reduction, or satisfaction scores (Mitchell et al., 2023; Nurchis et al., 2025). Prospective validation at KNUST University Hospital is required to quantify minutes saved and to confirm safe nurse override rates. Training data class imbalance may under-weight rare urgent phrasing; deployment should treat low-confidence or contradictory presentations as mandatory nurse review (Berkowitz et al., 2019; South African Triage Group, 2012). SMOTE increases Red recall but introduces false Red alerts (precision 96.41%); the baseline checkpoint is the project default unless clinical stakeholders prioritise zero missed Red cases. Holdout accuracy on the 80k corpus is an upper bound on offline label agreement, not a guarantee at live intake (Section 4.2.2).

CHAPTER 5

CONCLUSION

5.1 Summary

This thesis developed a mathematical pipeline from unstructured chief complaint text to South African Triage Scale (SATS) colour and clinical pathway at hospital intake. The chatbot accepts a single English free-text submission and returns acuity probabilities, a colour, and a pathway card aligned with SATS and KNUST University Hospital department structure.

Methodologically, the work specifies tokenisation, contextual embeddings, twelve-layer BioBERT encoding with multi-head self-attention, softmax classification, the deterministic map f_{SATS}, a clinical relevance gate, and department routing across seventeen hospital units. Training is documented as a three-stage stack: PubMed masked language modelling (Stage 1), encoder transfer from a prior triage checkpoint (Stage 2), and comprehensive project fine-tuning on the Triagegeist English benchmark (80,000 pairs, Section 3.1.2) with a reinitialised five-class head (Stage 3).

Holdout evaluation reports 99.92% accuracy, 0.9977 macro-F1, and 98.14% Level 1 recall on 8,000 validation complaints, with approximately 100 ms inference latency (Chapter 4, Figures 4.9–4.7). A full five-scenario pathway map (gate reject, Red, Orange, Yellow, Green) routes each NLP outcome to clinical and support departments at KNUST University Hospital, addressing intake delay, queue congestion, and acuity discontinuity described in Chapter 1.

5.2 Limitations and future work

Data scope. Reported holdout metrics come exclusively from the Triagegeist synthetic English benchmark (Section 3.1.2); the model is English-only at intake. A site visit to KNUST University Hospital informed pathway design and generated supplementary English example complaints for prototype and qualitative analysis, but those examples were not used for fine-tuning. A natural next step is prospective collection of nurse-labeled chief complaints at KNUST University Hospital for local retraining, and exploring optional fusion of vitals and TEWS alongside free text when measurements become available.

Class imbalance. Chapter 4 already evaluates baseline and SMOTE-augmented checkpoints with stratified holdout metrics. Future work can refine minority-class sensitivity for urgent levels while preserving specificity on routine presentations, building on the imbalance-mitigation pipeline documented in Section 4.2.

Clinical validation. Holdout evaluation establishes strong offline agreement with nurse labels. The next phase is a prospective pilot at KNUST University Hospital to measure waiting times, nurse override rates, and patient outcomes in live intake, converting offline performance into operational evidence (Mitchell et al., 2023).

Pathway refinement. The five-scenario pathway map is SATS-aligned and complete for gate reject, Red, Orange, Yellow, and Green routing. Ongoing collaboration with hospital administrators and clinical leads at KNUST University Hospital will keep department routing tables current as service lines evolve (Rominski et al., 2014; South African Triage Group, 2012).

System integration. The prototype `/predict` API and pathway cards already anchor the chatbot in a deployable pipeline. Deeper integration with Hospital Information Technology and Health Information systems would automate timer alerts and bed assignment, completing the Flow Management Engine vision across Administration, Diagnostics, and all clinical departments listed in Table 3.22.

REFERENCES

Berkowitz, A. L. et al. (2019). Under-triage: A new trigger to drive quality improvement in the emergency department. *Emergency Medicine Journal*.

Chawla, N. V., Bowyer, K. W., Hall, L. O., and Kegelmeyer, W. P. (2002). Smote: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357.

Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*.

Gligorijević, V. et al. (2018). Deep attention model for triage of emergency department patients. *arXiv preprint arXiv:1804.03240*.

KATH Emergency Department (2018). The implementation of the south african triage scale at komfo anokye teaching hospital. *Local implementation report*. See papers folder in project repository.

Laitinen-Fredriksson Foundation (2025). Triagegeist: Ai in emergency triage. Kaggle competition dataset. Synthetic emergency-department benchmark; `train.csv` and `chief_complaints.csv`.

Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H., and Kang, J. (2020). Biobert: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4):1234–1240.

Mitchell, R., O'Reilly, G., Mitra, B., et al. (2023). Systematic review: What is the impact of triage implementation on clinical outcomes in LMICs? *Academic Emergency Medicine*.

Nurchis, M. C. et al. (2025). Assessing the impact of waiting time on triage color code assignment and one-year mortality. *Health Science Reports*.

Rominski, S. D., Luboga, S., Mould-Millman, N.-K., et al. (2014). The implementation of the South African Triage Score (SATS) in an urban teaching hospital, Ghana. *African Journal of Emergency Medicine*, 4:71–75.

South African Triage Group (2012). The south african triage scale (sats) training manual.

Stewart, J., Lu, J., Goudie, A., et al. (2023). Applications of natural language processing at emergency department triage: A narrative review. *PLOS ONE*, 18(12):e0279953.

Sundström, B. W. et al. (2014). A pathway care model allowing low-risk patients to gain direct admission to a hospital medical ward. *Scandinavian Journal of Trauma, Resuscitation and Emergency Medicine*, 22:72.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems*.

Yilmaz, E. (2021). The reasons of apply to the emergency department by priority 3 (green tags) coded patients. *South Clin Ist Euras*, 32(2).

APPENDIX

5.3 Software repository

The implemented prototype and fine-tuned BioBERT artefacts are maintained at:
`https://github.com/samuel6814/triage`
No full source code is reproduced in this thesis; the mathematical specification in Chapter 3 and the system architecture in Figure 3.12 are the primary references.